TOOLKIT

Master Glossary

Every new word from every day, A to Z.

Every new word from every day, A to Z. The tag (e.g. `W1D1`) shows the day it was introduced. This file is rebuilt automatically from the daily glossaries.

0️⃣ Lag 0

- **Plain English:** The first bar of every correlogram. Always exactly 1, because a series matches itself perfectly.
- **Naija analogy:** Asking "do you look like yourself?" Of course you do. Ignore that question.
- **Example:** The grey bar at the far left of every correlogram today.

`W2D2`

✖️ 1.4826

- **Plain English:** The number that converts MAD into the same size as a standard deviation, so "3 robust σ" means what you expect.
- **Naija analogy:** The exchange rate between two currencies, so both prices mean the same thing.
- **Example:** `robust_sigma = MAD * 1.4826`

`W3D2`

1️⃣ Lag-1 autocorrelation

- **Plain English:** How much today matches yesterday.
- **Naija analogy:** If it rained heavily today, there's a good chance it rains tomorrow too in July.
- **Example:** `s.autocorr(lag=1)` gives 0.58 for throughput.

`W2D2`

1️⃣ One-step-ahead forecast

- **Plain English:** A forecast for just the next period, made with the real value of the period before it.
- **Naija analogy:** Predicting tomorrow's bus fare when you already know today's.
- **Example:** The regression used last week's real count every week: a one-step forecast. One-step naive scored 6.75 on the same weeks.

`W5D3` `W6D1`

3️⃣ 3-sigma rule

- **Plain English:** On normal data, only about 0.27% of readings fall outside ±3σ by chance.
- **Naija analogy:** Someone who is never more than 15 minutes late suddenly arriving 2 hours late: that's not chance, something happened.
- **Example:** About 8 chance alarms in 2,880 normal hours.

W3D2

🎯 3σ band (three-sigma)

- **Plain English:** Average ± 3 standard deviations. Readings outside it are rare if everything is normal.
- **Naija analogy:** "Normal fare is ₦500, give or take ₦100". ₦1,000 means something is up.
- **Example:** Sunlit 3σ band: 26.43 to 29.99 A. Only the 4 dropout hours fall outside.

W3D1

3️⃣ Three-cycle rule

- **Plain English:** Have at least three full cycles of data before you trust a seasonal pattern.
- **Naija analogy:** One friend says the new buka is sweet: gist. Three different friends: pattern.
- **Example:** 36 monthly rows = exactly 3 rainy seasons: the minimum.

W2D1

🗣️ 4-sentence finding

- **Plain English:** Situation → Observation → Implication → Recommendation. A whole report in four sentences.
- **Naija analogy:** How a good mechanic talks: "Your car is old (S). The brake pads are finished (O). You could crash (I). Change them today (R)."
- **Example:** The last slide of today's deck. The recommendation always comes last.

W1D2

🤦🏾 Accuracy (and why it lies)

- **Plain English:** The fraction of all readings a detector got "right". Useless when real events are rare.
- **Naija analogy:** Ade the guard sleeps all night and is "97% accurate", because most nights there's no thief.
- **Example:** A detector that never fires scores 99.87% on last week's 4,706 readings.

W4D1

➕ Additive model

- **Plain English:** observed = trend + seasonal + residual. The seasonal swing stays the same size even when the level rises.
- **Naija analogy:** December transport fare goes up by a fixed ₦100 every year, whatever the normal fare is.
- **Example:** `seasonal_decompose(series, model="additive", period=12)`

W2D1

🤖 AI button (Gemini)

- **Plain English:** Colab's built-in AI helper. Great for explaining code and errors.
- **Naija analogy:** A very smart friend who talks a lot. Never tell him company secrets.
- **Example:** OK to ask: “Explain this error.” Never paste real NIGCOMSAT telemetry.

W1D1

🤖 AI-use note

- **Plain English:** One paragraph in your report saying what you used an AI assistant for, how you checked its output, and how you kept restricted data out of it.
- **Naija analogy:** Declaring what's in your luggage at the airport: honest and short, and everyone relaxes.
- **Example:** "I described the column names and the shape of the export; I never pasted rows. I tested every function on the course data first."

W6D3

😴 Alarm fatigue

- **Plain English:** When there are so many false alarms that people stop paying attention, even to the real ones.
- **Naija analogy:** After the 10th "thief!" shout from the mai-guard, nobody comes out anymore.
- **Example:** A NOC that sees 46 useless solar-array alarms a season learns to ignore that channel.

W3D1 W4D1

🔔 Alarm (limit check)

- **Plain English:** A rule that shouts when a reading crosses a fixed number.
- **Naija analogy:** The smoke detector in the kitchen: it doesn't know you're frying plantain, it just sees smoke.
- **Example:** `bus["solar_array_current_a"] < 20` fires an alarm.

W3D1

🚨 Alert line (threshold)

- **Plain English:** A value that, once crossed, means someone should act. Real ones are set by engineers.
- **Naija analogy:** The fuel light on your car's dashboard.
- **Example:** The lab's 97 Ah line is **made up for practice**.

W2D3

🎚️ Alpha

- **Plain English:** The setting for exponential smoothing. High alpha = follows the data closely and reacts fast. Low alpha = smooth and slow.
- **Naija analogy:** High alpha believes every WhatsApp broadcast immediately. Low alpha waits for three people to confirm.
- **Example:** alpha 0.5 hugs the daily line; alpha 0.05 is very calm.

W1D3 W5D2

🚨 Anomaly

- **Plain English:** A reading that doesn't fit what normally happens.
- **Naija analogy:** Your quiet neighbour suddenly playing loud music at 3am. Something is off.
- **Example:** The solar-array current suddenly dropping to about half while the sun is shining.

W3D1

🔑 Answer key (labels / ground truth)

- **Plain English:** A list of the real events: what happened, where, and when. It turns "my detector seems fine" into a score.
- **Naija analogy:** The WAEC marking scheme. Without it, every student says "I passed".
- **Example:** `telemetry_anomaly_labels.csv`: 6 events, EV01 to EV06.

W4D1

🔪 .asfreq()

- **Plain English:** Declares the rhythm of your data (e.g. daily) and creates an empty slot for any missing period.
- **Naija analogy:** A class register with every school day printed on it. An absent student shows as a blank, not a missing page.
- **Example:** `ka.asfreq("D")`. Missing days appear as `NaN`.

W1D1

✅ assert

- **Plain English:** A line that checks something is true and stops the notebook with an error if it isn't. A guard that fails loudly.
- **Naija analogy:** The gateman who refuses entry if your name isn't on the list. No argument.
- **Example:** `assert bus.index[train].max() < bus.index[test].min()` proves the split has no overlap.

W5D1

🧠 Autocorrelation

- **Plain English:** How much a series looks like its own shifted copy. One number from −1 to +1.
- **Naija analogy:** How much your Monday looks like last Monday. Same traffic, same stress, same suya spot.
- **Example:** Throughput: lag 1 = 0.58, lag 7 = 0.99.

W2D2

🙈 Automatic leak test

- **Plain English:** A computer check of "could I have known this feature in time?": hide this week's answer and the whole future, rebuild the feature, and see if it changes. If it changes, it leaked.
- **Naija analogy:** Collecting the phones before the exam starts. If your answer changes when the phones are gone, it wasn't your answer.
- **Example:** `shift(1)` passes; `shift(-1)`, a centred window and `rolling(4).mean()` without a shift all fail.

W6D3

✅ Availability

- **Plain English:** The percentage of time a link worked properly.
- **Naija analogy:** Out of 24 hours, how many did your network actually give you full bars?
- **Example:** 99.6% availability = about 6 minutes of bad service in a day.

W1D1

🔁 Backtesting

- **Plain English:** Replaying a forecasting method on the past, many times, using only what was known at each moment, and scoring it against what really happened next.
- **Naija analogy:** Before you pay Oga Tipster for VIP tips, your cousin checks his tips against every match from last season.
- **Example:** The Week 5 Day 3 regression replayed on six 4-week stretches of 2025, refitted each time.

W6D1

🔁 Backtesting the method

- **Plain English:** Repeating the whole forecasting recipe as if you stood at several earlier dates, with only the data you had then, and checking each call against what happened.
- **Naija analogy:** Testing your football-prediction system on the last four seasons, not just last Saturday.
- **Example:** Battery calls made from 1, 2, 3 and 4 years of data: −74, −37, −13 and +4 days against the first dip.

W6D2

📏 Baseline

- **Plain English:** The simplest reasonable method, the one every fancier method must beat on the same test.
- **Naija analogy:** Grandma's weather forecast: “tomorrow will be like today”. Beat her or go home. 👵🏾
- **Example:** “Same weekday last week” as a forecast for throughput.

W2D3 W5D1 W5D2

👵🏾 Baseline forecast

- **Plain English:** The simplest possible forecast. Any clever model must beat it, or it's not worth using.
- **Naija analogy:** Grandma's weather forecast: "tomorrow will be like today." Hard to beat!
- **Example:** "Same as yesterday" and "same weekday last week".

W2D2

🙈 Blind forecast

- **Plain English:** Every week of the horizon is forecast from the origin, without using any real value that arrives later.
- **Naija analogy:** Writing your full betting slip before the first match kicks off. No changing it at half-time.
- **Example:** The forecast for 22 December can't use the real count from 15 December.

W5D2

🙈 Blind (multi-step) forecast

- **Plain English:** Forecasting several periods ahead from one starting point, without seeing any of the real values in between.
- **Naija analogy:** Planning the whole month's foodstuff on the first of the month.
- **Example:** Naive, seasonal naive and the rate seasonal naive all forecast 3 Nov – 22 Dec blind, standing on 27 Oct.

W5D3 W6D1

🎯 Brief

- **Plain English:** One of the five project types you can choose (A–E), each with its own operational question.
- **Naija analogy:** The topic the lecturer approves for your final-year project.
- **Example:** Brief B: "When will the battery be too tired to carry us through eclipse?"

W2D3

📶 C/N (carrier-to-noise ratio)

- **Plain English:** How strong the signal is compared to the background hiss, in dB. Higher is better.
- **Naija analogy:** How clearly you can hear your friend on the phone in a noisy bus park.
- **Example:** About 14.2 dB in clear sky; it falls in heavy rain.

W3D1

🥁 Cadence (frequency)

- **Plain English:** How often a reading is recorded: hourly, daily, weekly, monthly.
- **Naija analogy:** How often your landlord comes for rent: monthly? yearly? (Hopefully not weekly 😅)
- **Example:** `bus_power_telemetry` is hourly; `ka_band_availability` is daily.

W1D1

🌧️ Calibration weeks

- **Plain English:** The past weeks whose misses you use to size a band. They must include the kind of trouble you'll be forecasting.
- **Naija analogy:** Testing a new umbrella in harmattan and declaring it rain-proof.
- **Example:** Misses from summer only (from 19 May) gave ± 3.33 and covered just 2 of 8 November–December weeks.

W5D3

🔋 Capacity (amp-hours, Ah)

- **Plain English:** How much charge a battery can still hold. It shrinks as the battery ages.
- **Naija analogy:** Your old phone that used to last all day and now dies before lunch.
- **Example:** `est_capacity_ah` falls from 103.3 Ah to 96.6 Ah over three years.

W2D3

✅ Catch (true positive)

- **Plain English:** A real event the detector flagged.
- **Naija analogy:** The mai-guard shouts and there really is a thief at the gate.
- **Example:** The sunlit control chart catches all 3 solar-array dropouts.

W4D1

🧱 Cell (code cell / text cell)

- **Plain English:** One block in a notebook. A **code cell** runs Python; a **text cell** is just for reading.
- **Naija analogy:** One stop on a danfo route. You can't jump to Oshodi without passing the stops before it.
- **Example:** `print("Hello")` lives in a code cell. These instructions live in text cells.

W1D1

➖ Centre line

- **Plain English:** The middle line of a control chart: the average of the quiet reference period.
- **Naija analogy:** The "normal" fare for your usual bus route.
- **Example:** 27.87 A for 1–15 August.

W3D2

🔮 Centred window

- **Plain English:** A window with today in the middle: half the days are in the past, half in the **future**.
- **Naija analogy:** Marking a WAEC script after you've seen the answer sheet. Looks perfect, but it's cheating if you use it live.
- **Example:** `s.rolling(30, center=True).mean()` is empty for the last 15 days.

W1D3

📍 Change point

- **Plain English:** The moment a series switches to a new behaviour: a new level or a new speed.
- **Naija analogy:** The day the fuel price changed at the pump. Before and after are two different worlds.
- **Example:** 12 May 2025 for the amplifier; 1 April 2025 for the battery.

W3D3

🚩 Checkpoint

- **Plain English:** A short progress check at a fixed point in the course.
- **Naija analogy:** Police checkpoints on the expressway, but these ones help you. 😄
- **Example:** Week 3 · Day 3: bring your unit head's reply and your profiled data.

W2D3

🧼 Clean residual

- **Plain English:** A residual you can trust: mean near zero, no autocorrelation left, and the same spread all year.
- **Naija analogy:** A clean bill of health from the doctor before you start the race.
- **Example:** After removing the 2 spike days, lag-1 autocorrelation fell from 0.33 to 0.06.

W3D2

🧩 Clustering

- **Plain English:** Grouping rows that look alike, without being told the groups. Useful for finding operating **modes**.
- **Naija analogy:** At a wedding, without any seating plan, the groom's people, the bride's people and the gate-crashers still end up at different tables.
- **Example:** k-means put every eclipse hour (92 of 92) in one cluster and every recharge hour (276 of 276) in another.

W4D2

⚖️ Coefficient

- **Plain English:** The weight a regression gives one column: how much the forecast moves when that column goes up by one, with the others held still.
- **Naija analogy:** The price per derica of garri: buy one more, pay that much more.
- **Example:** `terminals` +0.0231 per terminal, about +23 tickets a week per 1,000 terminals. Positive, as it should be.

W5D3

🚨 Coefficient sanity check

- **Plain English:** Reading each coefficient's sign (+ or −) and asking whether it makes physical sense. A sign that makes no sense means something is broken until proven otherwise.
- **Naija analogy:** If the receipt says the more fuel you buy, the less you pay, check the receipt before celebrating.
- **Example:** A negative terminals coefficient would say "bigger network, fewer tickets": investigate.

W5D3

🌊 Common-cause variation

- **Plain English:** The everyday, built-in wobble of a stable process. Nothing to fix.
- **Naija analogy:** Lagos traffic being a bit slower on Tuesday than Monday.
- **Example:** Array current wobbling between 27.8 and 28.0 A.

W3D2

🧩 Component

- **Plain English:** One of the parts a decomposition produces.
- **Naija analogy:** One instrument in a highlife band. The song is all of them playing together.
- **Example:** `parts.trend`, `parts.seasonal`, `parts.resid`

W2D1

🌞 Conditioning on a known cause

- **Plain English:** Only judging readings where a known, normal cause is absent, instead of widening the alarm band for everyone.
- **Naija analogy:** You don't call an electrician when the lights go off during a scheduled NEPA load-shedding.
- **Example:** Adding `eclipse_flag == 0` took the control chart from 46 false alarms to 0.

W4D1

🟦 Confidence band

- **Plain English:** The shaded area in a correlogram. Bars inside it could easily be chance.
- **Naija analogy:** The "e no too matter" zone: small differences you shouldn't argue about.
- **Example:** For 730 days of data the band is about ±0.07.

W2D2

🪦 Contamination

- **Plain English:** The share of training rows you **allow** the forest to flag. It sets the line on the score.
- **Naija analogy:** Telling the gateman "you can stop at most 5 cars in every 1,000 for searching".
- **Example:** `contamination=0.005` (0.5%) on ~8,760 hours ≈ 44 flags expected. We got 58: same ballpark.

W4D2

🎭 Context (state-based limits)

- **Plain English:** Checking which mode the system is in before deciding whether a reading is strange.
- **Naija analogy:** A dark house at midnight is normal. A dark house at noon with the gen running is not.
- **Example:** `floor_alarm & (bus["eclipse_flag"] == 0)` only checks the limit in sunlight.

W3D1

🔪 Control chart

- **Plain English:** A chart with a centre line and two limits that shows when a reading leaves its normal range.
- **Naija analogy:** Mama Put's price sense: she knows the usual range for a basket of tomatoes, so one crazy price jumps out.
- **Example:** Solar-array current with a band from 27.65 A to 28.09 A.

W3D2

↕️ Control limits (UCL / LCL)

- **Plain English:** The upper and lower edges of "normal": centre ± 3σ. Outside them = unusual.
- **Naija analogy:** The fare range where you don't argue. Above it, you ask the conductor, "Ehn?!"
- **Example:** LCL 27.65 A, UCL 28.09 A.

W3D2

🔗 Correlation

- **Plain English:** A number from −1 to +1 showing how much two series move together. It does **not** tell you which one causes the other.
- **Naija analogy:** Umbrella sales and traffic jams both rise when it rains. Umbrellas don't cause traffic.
- **Example:** Rain and uplink power: correlation 0.97.

W1D2

💔 Correlation break

- **Plain English:** Two sensors that normally move together stop doing so, even though each value is inside its own normal range.
- **Naija analogy:** Husband and wife who always arrive at church together. One day only one comes, smiling. Nothing is wrong with either face; something is wrong.
- **Example:** EV02 and EV05: sunlight is normal, yet the battery stops charging.

W4D3

🧭 Correlation ≠ cause

- **Plain English:** Moving together doesn't prove one causes the other. Even when there IS a cause, check which direction it runs.
- **Naija analogy:** Umbrella sales and rain go together. Selling umbrellas doesn't bring rain.
- **Example:** Rain and uplink power: r = 0.97, because rain makes the station turn up its power, not the other way round.

W2D2

🔗 Correlation (r)

- **Plain English:** How closely two different series move together, from −1 to +1.
- **Naija analogy:** How often Mama Ngozi and Mama Bisi show up at the same parties.
- **Example:** `a.corr(b)`

W2D2

📊 Correlogram (ACF plot)

- **Plain English:** A bar chart with one bar per lag, showing the autocorrelation at each.
- **Naija analogy:** A report card for memory: one score for yesterday, one for last week, one for last month.
- **Example:** `plot_acf(s.dropna(), lags=35)`.

W2D2

🎯 Coverage

- **Plain English:** The share of held-out periods whose real value actually landed inside the band. It's how you check a band's promise.
- **Naija analogy:** The weatherman who says "90% chance of rain": count how often it actually rained.
- **Example:** The code book's band covered 5 of 8 weeks (62.5%): too narrow for a "90%" claim.

W5D3

⏰ Cron

- **Plain English:** The Linux timetable for scheduled jobs. A line has five time fields (minute, hour, day of month, month, day of week) and then the command. `*` means "every".
- **Naija analogy:** The church bell that rings at 6 a.m. every day. Nobody presses it; it's on the timetable.
- **Example:** `0 7 * * *` = 07:00, every day.

W6D3

🪟 Crossing window

- **Plain English:** For a noisy reading, "the day it crossed" isn't one day: readings bounce either side of the limit for a while. Report the window.
- **Naija analogy:** "When did harmattan start?" Some mornings are dusty in November, but you only stop arguing in December.
- **Example:** The battery's first daily dip below 80% was 14 May 2025; it was last at or above 80% on 24 Jul 2025. 49 of the 72 days in between were below.

W6D2

📄 CSV file

- **Plain English:** A plain-text table where commas separate the columns.
- **Naija analogy:** A list written in a jotter with commas instead of lines.
- **Example:** `date,tickets_raised` / `2024-01-01,8`

W1D1

🧾 CUSUM (cumulative sum)

- **Plain English:** A detector that keeps a running total of small leftovers and alarms when the total gets too big.
- **Naija analogy:** Your landlord's "small-small" rent increases. No single year shocks you, but add up the receipts and you've paid ₦415k extra.
- **Example:** `S = max(0, S + leftover − k)` caught the amplifier's step on 12 May 2025.

W3D3

📈 Cycle

- **Plain English:** Ups and downs that repeat, but **with no fixed length**.
- **Naija analogy:** Business boom years and slow years. Nobody can tell you the exact date of the next boom.
- **Example:** Demand strong for 2 years, weak for 3, then rising again.

W1D1

〰️ Daily average (mean)

- **Plain English:** Add up all the readings in a day and divide by how many. Shows the typical level.
- **Naija analogy:** "On average, NEPA gave us 20 hours today", which hides that the 4 missing hours were during the match.
- **Example:** `rf["link_margin_db"].resample("D").mean()` : never below 4.9 dB in August.

W1D2

📉 Daily minimum

- **Plain English:** The single lowest reading of each day: the worst moment.
- **Naija analogy:** The one hour you sat on Third Mainland Bridge: that's the hour you remember.
- **Example:** `resample("D").min()` showed an outage on 11 of 31 days.

W1D2

📋 DataFrame

- **Plain English:** pandas' name for a table: rows and named columns.
- **Naija analogy:** The class register: one row per student, one column per day.
- **Example:** `ka` is a DataFrame with 1,096 rows.

W1D1

📉 dB (decibel)

- **Plain English:** A unit for signal strength on a "times" scale: +3 dB is about double, −3 dB about half.
- **Naija analogy:** Like Richter scale for earthquakes: small number changes mean big real changes.
- **Example:** The worst hour in August hit −4.3 dB of margin.

W1D2

🧰 De-orbit reserve

- **Plain English:** Propellant kept back so that at end of life the satellite can be moved out of the busy geostationary ring. It's a limit, not spare fuel.
- **Naija analogy:** The transport fare you never spend at the party, so you can get home.
- **Example:** 5 kg, from `degradation_thresholds.csv`.

W6D2

🦕 Decelerating fade

- **Plain English:** The loss rate itself gets smaller over time, so the trend is a gentle curve. A straight line fitted on early years then overstates the later fade and calls the crossing early.
- **Naija analogy:** A new car loses value fast in the first years, then more slowly.
- **Example:** Battery fade per year: −4.81, −4.69, −4.55, −4.45, −4.34%.

W6D2

📉 decision_function (anomaly score)

- **Plain English:** A number for every row saying how odd it is. **Lower = more odd.** `predict` is just this score compared with a line.
- **Naija analogy:** Instead of "pass/fail", the full exam mark. You can rank everyone by it.
- **Example:** `score.nsmallest(10)` : the ten oddest hours of the year.

W4D2

🚨 Decision limit (h)

- **Plain English:** How big the running total must get before CUSUM raises the alarm.
- **Naija analogy:** How many times your neighbour's generator must wake you before you go and knock.
- **Example:** `h = 5 × σ` = 1.02 °C. Lower h = faster but noisier.

W3D3

🔧 Decomposition

- **Plain English:** Splitting one time series into separate parts: trend, seasonal and residual.
- **Naija analogy:** Tasting the jollof and naming each ingredient: rice, tomato base, and that one pepper that surprised you.
- **Example:** `seasonal_decompose(kam["availability_pct"], period=12)`

W2D1

📉 Degradation

- **Plain English:** The slow, one-way loss of performance as equipment ages: capacity, gain or fuel that won't come back.
- **Naija analogy:** Your phone battery: brand new it lasts two days, after three years it begs for the charger by afternoon.
- **Example:** Battery capacity fell from 99.86% to 77.84% over five years.

W6D2

🎤 Demo

- **Plain English:** A short, live showing of your tool to the people who will use it: what problem, what data, what method, how you tested it, the tool running, and what to do next.
- **Naija analogy:** The owambe MC gives the groom's friends five minutes. When the time is up, the DJ takes over, ready or not.
- **Example:** Six beats in 5:00: problem 0:45, data 0:45, method 0:45, honest test 1:00, the tool live 1:00, recommendation 0:45.

W6D3

🚀 Deployment

- **Plain English:** Putting your tool where its users can actually use it, every day, without you. For this course: on the **training server**, as a dashboard or a scheduled report.
- **Naija analogy:** Building the borehole is one thing. Deployment is when the whole street can fetch water without calling you.
- **Example:** A tool that only runs on your laptop earns at most half of the deployment marks.

W6D3

🪫 Depth of discharge

- **Plain English:** How far the battery was drained, as a percentage of what it holds.
- **Naija analogy:** How low your inverter battery went during last night's blackout.
- **Example:** Under 1% on normal days, up to about 22% on the deepest eclipse days.

W2D3

🪨 Design life

- **Plain English:** How long the manufacturer built a satellite to operate. Real satellites can outlive it or fall short; remaining-useful-life analysis tells you which way it's going.
- **Naija analogy:** "This generator is rated for 10 years", but how you use it decides the real answer.
- **Example:** NigComSat-1R, launched December 2011, has a 15-year design life.

W6D2

⏳ Detection delay

- **Plain English:** How many days pass between the onset and the alarm.
- **Naija analogy:** The time between “I don pay” and the bank alert landing.
- **Example:** 3 days for the practice fault with $h = 5\sigma$; 6 days with $h = 10\sigma$.

W3D3 W4D1

🧪 Detector (model)

- **Plain English:** Any rule or method that decides "normal" or "not normal". Even a fixed limit is a simple one.
- **Naija analogy:** A gateman deciding who looks suspicious. Some gatemen are better than others.
- **Example:** The 20 A floor is a detector. So is mean − 3σ.

W3D1

➖ Differencing (.diff())

- **Plain English:** Replacing each value with the change since the previous step. It removes the trend.
- **Naija analogy:** Instead of asking "how much money do you have?", asking "how much did you gain or lose this week?"
- **Example:** Suya vs generators: 0.97 on levels, 0.06 on changes.

W2D2

🦕 Drift

- **Plain English:** A slow, steady slide. No single day looks strange.
- **Naija analogy:** Your eyesight getting a little worse every year until one day you can't read the danfo number.
- **Example:** Battery capacity fading from 103 to 95.4 Ah over three years.

W3D1

📡 Driver (exogenous variable)

- **Plain English:** A second series that pushes the one you forecast, and that you know at the moment you forecast.
- **Naija analogy:** The number of guests drives how much rice you cook.
- **Example:** Active VSAT terminals drive the number of tickets.

W5D2

☝🏾 Dual axis (avoid it)

- **Plain English:** Two different y-axes on one chart, one on each side. It lets you make any two lines look related.
- **Naija analogy:** Two referees on one pitch with different rule books.
- **Example:** Rain and uplink power go in two stacked charts instead.

W1D2

👯 Duplicate timestamp

- **Plain English:** The same time appearing twice: usually an export mistake.
- **Naija analogy:** Being charged twice on one POS transaction.
- **Example:** Two rows both saying `2024-03-06`.

W1D1

👨🏾‍💻 Duty engineer / page

- **Plain English:** The engineer on shift who gets alerted (paged) when an alarm fires, and must check it.
- **Naija analogy:** The one who gets the 2 a.m. "Oga, come and see" phone call.
- **Example:** Every false alarm is one page to the duty engineer.

W4D1

🌒 Eclipse

- **Plain English:** The hour when the Earth's shadow covers the satellite, so the solar panels give no power and the battery takes over.
- **Naija analogy:** The satellite's own "NEPA has taken light". The battery is its inverter.
- **Example:** 46 eclipse hours in this file, every one at 21:00 UTC (22:00 in Lagos).

W3D1

🌒 Eclipse season

- **Plain English:** The weeks around each equinox (March and September) when the Earth's shadow crosses the satellite's orbit for up to 72 minutes a day.
- **Naija analogy:** NEPA's scheduled load-shedding season, but for the satellite. It switches to battery.
- **Example:** Eclipse hours sit at 21:00 UTC from 1 September to 16 October in the bus-power file.

W2D1 W2D3 W6D2

🪑 Embargo

- **Plain English:** A gap left out between training and test, at least as long as your longest feature window, so no feature on the first test hour can reach back into training.
- **Naija analogy:** The empty seat between two candidates in the exam hall.
- **Example:** 1 September (24 hours) is in neither set, because W4D3's features use 24-hour windows.

W5D1

🔋 End-of-discharge voltage

- **Plain English:** The battery's voltage at the end of an eclipse, when it has worked hardest. Lower means a deeper discharge or an older battery.
- **Naija analogy:** How much power is left in your inverter battery just before NEPA brings light back.
- **Example:** `eod_voltage_v` dips in March and September and slowly drifts down over the years.

W2D1 W2D3

🌍 Equinox

- **Plain English:** The two days a year (around 21 March and 23 September) when day and night are equal. Eclipse seasons are centred on them.
- **Naija analogy:** The two fixed "owambe Saturdays" of the satellite calendar: you can plan for them years ahead.
- **Example:** The seasonal bumps in the lab land in March and September.

W2D3

📌 Event

- **Plain English:** One real problem with a start and an end. It can last one hour or many months.
- **Naija analogy:** One power outage. Whether NEPA takes light for 5 minutes or 5 days, it's one outage.
- **Example:** EV05, the amplifier's cooling-loop blockage, lasts 233 days.

W4D1

🎯 Event-level scoring

- **Plain English:** Scoring per event (caught or missed), not per row. A 233-day event is one thing to catch.
- **Naija analogy:** A teacher marks "did you answer question 4?", not "how many lines did you write for question 4?".
- **Example:** Flagging one day of EV05 counts as one catch, not 233.

W4D1

🧱 Expected value (model)

- **Plain English:** What a reading *should* be, given what we already know (drive level, time of year…).
- **Naija analogy:** Mama Put expects tomatoes to be dear in the scarce season. That's her model.
- **Example:** On 3 Sep 2024 the recipe expected 63.1 °C.

W3D2

🕴🏾 Expo (data leakage)

- **Plain English:** When information from the test sneaks into training or into your choices, so the score looks better than the model really is. Data scientists call it **leakage**.
- **Naija analogy:** WAEC "expo": you saw the questions the night before, so your A1 measures the expo, not you.
- **Example:** The copycat found 100% of fault hours with a random split, and 3.4% when split by time.

W5D1

🫖 Exponential smoothing

- **Plain English:** A running "level" that each new value pulls part of the way towards itself. Recent weeks count most, old weeks fade away.
- **Naija analogy:** How your opinion of a bus driver updates: today's ride counts a lot, last year's ride a little.
- **Example:** With α = 0.3 the final level is 75.6. MAE 12.80.

W5D2

📰 Exponential smoothing (`.ewm`)

- **Plain English:** A smoother that gives the newest values the most weight and older values less and less.
- **Naija analogy:** How you rate a footballer: last week's match counts most, matches from last season count a little.
- **Example:** `s.ewm(alpha=0.3).mean()`

W1D3

🔭 Extrapolation

- **Plain English:** Using a trend beyond the range of data it was fitted on. The further you reach, the less you should trust it.
- **Naija analogy:** "He scored in his first two matches, so he'll score 38 this season." Maybe. Maybe not.
- **Example:** The amplifier call reaches 9.3 years past the three years it was fitted on.

W6D2

📉 Fade rate

- **Plain English:** How much capacity a battery loses per year.
- **Naija analogy:** How fast your new phone's battery stops lasting the whole day.
- **Example:** −2.24 Ah/yr before April 2025, −3.68 after.

W3D3

Fair comparison

- **Plain English:** Two detectors judged on the same held-out months, the same hours and the same scorer, with settings fixed before the test. Anything else is an opinion.
- **Naija analogy:** Two strikers, same match, same referee, same goalposts.
- **Example:** The forest and the chart, both on steady hours of Sep–Dec, both through `score_events`.

W5D1

False alarm

- **Plain English:** An alarm when nothing is actually wrong.
- **Naija analogy:** The boy who cried wolf. Or the car alarm that goes off for every keke.
- **Example:** The 46 eclipse-night alarms from the 20 A floor.

W3D1

False-alarm budget

- **Plain English:** How many pointless alarms you can afford before people stop listening.
- **Naija analogy:** The car alarm on your street that goes off for every keke. After a week, nobody even looks.
- **Example:** ±2σ ≈ 131 false alarms per 2,880 hours; ±3σ ≈ 8.

W3D2

False alarm (false positive)

- **Plain English:** A flag when nothing was really wrong. Each one costs someone's time.
- **Naija analogy:** Musa shouting "Thief!" at the neighbour's goat. Again.
- **Example:** The SoC floor raised 18 false alarms, all on normal eclipse nights.

W4D1

Feature

- **Plain English:** One column of the table you give a model. It can be a raw sensor reading or something you compute from the readings.
- **Naija analogy:** The questions a gateman asks before he opens the gate: who are you, who sent you, did you come yesterday? Each question is a feature.
- **Example:** `battery_soc_pct` is a raw feature; `temp_per_drive` is one we built.

W4D3

🛠️ Feature engineering

- **Plain English:** Building new columns so the model can see what an engineer already knows: history, operating state, and how sensors relate.
- **Naija analogy:** Teaching a new apprentice mechanic *what to listen for*. The engine noise was always there; now he knows which knock matters.
- **Example:** Dividing amplifier temperature by RF drive turned four invisible weeks of EV04 into 605 flagged hours.

W4D3

📝 Finding (4-sentence)

- **Plain English:** A short write-up in four sentences: Situation, Observation, Implication, Recommendation.
- **Naija analogy:** How a good doctor explains: what's happening, what the test shows, what it means, what to do.
- **Example:** “Weekends average 3.4 tickets a day vs 7.1 on weekdays” is an Observation.

W2D3

🖼️ Finding title

- **Plain English:** A chart title that states what the chart shows, ideally with a number.
- **Naija analogy:** A newspaper headline: “Eagles win 2–0”, not “Football match”.
- **Example:** “Ka-band link went down on 11 of 31 August days”.

W1D2

➖ First difference (`diff`)

- **Plain English:** This hour's value minus the previous hour's: how much it just changed.
- **Naija analogy:** Not "how much fuel is in the tank?" but "how much did it drop since morning?"
- **Example:** `soc.diff(1)` jumps to about −7 at 01:00 on 2 May, when EV02 begins.

W4D3

💧 First dip

- **Plain English:** The first single reading below the limit. Early warning, but noise can trigger it.
- **Naija analogy:** The first raindrop of the season: real, but not yet the rainy season.
- **Example:** 14 May 2025. The 3-year call (1 May) is 13 days early against it.

W6D2

🏋️ fit

- **Plain English:** Training: the model studies the rows you give it and learns what is normal.
- **Naija analogy:** The new apprentice spends his first month just watching the master mechanic.
- **Example:** `model.fit(X[train])` learns from January–August only.

W4D2

🧩 Fold

- **Plain English:** One round of a backtest: one training block and the test weeks scored after it.
- **Naija analogy:** One match day in the season. A season has many.
- **Example:** Fold 5 trains on 42 weeks and tests 20 Oct – 10 Nov 2025.

W6D1

🔮 Forecast

- **Plain English:** A guess about a value that hasn't happened yet, made **only** from what was known at the time.
- **Naija analogy:** Mama Put deciding how much rice to cook tomorrow from how many plates she sold this week.
- **Example:** Standing on 27 Oct 2025, guess the NOC tickets for each of the next 8 weeks.

W5D2

❌ Forecast error

- **Plain English:** Actual minus forecast for one step. Positive = forecast too low, negative = too high.
- **Naija analogy:** You budgeted ₦5,000 for fuel and spent ₦7,000: an error of ₦2,000.
- **Example:** Week of 3 Nov: actual 53, naive 74, error −21.

W5D2

🔭 Forecast horizon

- **Plain English:** How far ahead a forecast reaches, here in weeks. Different questions need different horizons.
- **Naija analogy:** "Will it rain this evening?" and "Will it rain at my December wedding?" are different forecasts.
- **Example:** Next week's shift = 1 week; the month-ahead roster = 4 weeks.

W6D1

📍 Forecast origin

- **Plain English:** The moment you stand at when you make the forecast. Everything after it is unknown.
- **Naija analogy:** The day you buy your ticket at the motor park: you don't yet know how the road will be.
- **Example:** Today's origin is the last training week, 27 Oct 2025.

W5D2

🧍 Forecast origin (rolling origin)

- **Plain English:** The week you are standing in when you make the forecast. Everything before it may be used; nothing after it. A rolling-origin backtest makes a forecast from every week in turn.
- **Naija analogy:** Standing at the bus stop: you can see the danfos that have passed, not the ones still coming.
- **Example:** `rolling_origin(tickets, forecasters, h=8)` stands at 45 origins, from 30 Dec 2024 to 3 Nov 2025.

W6D1

📝 Four-sentence recommendation

- **Plain English:** The plain-language verdict for a manager: what to use, for which question, what not to use, and when to be careful.
- **Naija analogy:** What your mechanic tells you after the test drive: "Use the car for town, not for Abuja. Don't trust the spare tyre. The brakes go soft in the rain."
- **Example:** Naive for next week; the rate forecast for the month-ahead roster; not the regression; expect misses when the rains start or stop.

W6D1

🧾 Fuel-gauge sum (sanity check)

- **Plain English:** The no-model version of the answer: what's left above the limit ÷ what's used per year. If it disagrees with the model, look for a bug.
- **Naija analogy:** Counting the money in your pocket before trusting the bank app.
- **Example:** (101.93 − 5) kg ÷ 9.62 kg a year = 10.1 years: about January 2036. The line said December 2035.

W6D2

🕵🏾 Gap

- **Plain English:** A period with no reading at all, e.g. a missing day.
- **Naija analogy:** The days in your diary when you were "too busy" to write. 🙈
- **Example:** In B5 we found 3 missing days in March 2024.

W1D1

🛰️ GEO (geostationary) satellite

- **Plain English:** A satellite about 36,000 km up that moves with the Earth's spin, so it stays over the same spot.
- **Naija analogy:** A security light on a high pole: it always shines on the same compound.
- **Example:** NigComSat-1R sits at 42.5° East.

W1D1

🗺️ Geopolitical zones

- **Plain English:** Nigeria's six regional groupings: North West, North East, North Central, South West, South East, South South.
- **Naija analogy:** The six "houses" at inter-house sports.
- **Example:** The `region` column in the VSAT file.

W1D2

💻 Google Colab

- **Plain English:** A free website where you write and run Python. Google's computer does the work; yours just shows the screen.
- **Naija analogy:** A cyber café computer that lives in your browser, never runs out of fuel and never asks for ₦200 per hour.
- **Example:** You open `W1D1-exercise.ipynb` in Colab and run it without installing anything.

W1D1

📝 Held-out period

- **Plain English:** The stretch of time kept aside for the test: no fitting, no tuning, no peeking until the very end.
- **Naija analogy:** The sealed envelope of exam papers that only opens in the exam hall.
- **Example:** 2 Sep – 31 Dec: 2,904 hours, with EV04 and EV05 inside.

W5D1

🕰️ History feature (lag, difference, rolling statistic)

- **Plain English:** A column that carries the past onto the current row: the value an hour ago (`shift(1)`), the change since last hour (`diff(1)`), the average of the last day (`rolling(24).mean()`).
- **Naija analogy:** The landlord's receipt book: today's payment makes sense only next to last month's.
- **Example:** The code book's builder makes 6 history columns (plus the raw value) for each of 4 sensors.

W4D3

🔒 Hold-out (test weeks)

- **Plain English:** The last stretch of the series, locked away until the forecasts are made, then used to score them.
- **Naija analogy:** The sealed exam envelope that only opens in the hall (Week 5 · Day 1's "no expo").
- **Example:** The last 8 of the 104 weeks.

W5D2

⚖️ Honest test (Checkpoint 4)

- **Plain English:** The final-project checkpoint where you bring a split by time and a table comparing your method with a simple baseline on the same period, with the same scorer.
- **Naija analogy:** A football friendly on neutral ground, same referee for both teams.
- **Example:** Today's table: each method, how far ahead, MAE, 90% band, coverage. The rate seasonal naive wins: say so.

W5D3

🔭 Horizon

- **Plain English:** How far ahead you forecast, counted in steps (here, weeks).
- **Naija analogy:** "I'll pay you back in 8 weeks." The horizon is 8 weeks.
- **Example:** `H = 8` : forecast the 8 weeks from 3 Nov to 22 Dec 2025.

W5D2

↕️ Hysteresis

- **Plain English:** Switching the alarm ON at one level and OFF at a different level, so it doesn't flicker on and off.
- **Naija analogy:** The air-conditioner that starts at 26 °C but only stops at 23 °C, instead of clicking on and off all night.
- **Example:** Alarm on when margin drops below 2 dB, cleared only when it rises above 3 dB.

W3D1

🏠 In-sample error (marking your own homework)

- **Plain English:** Misses measured on the same weeks the model was fitted to. They are always flattering.
- **Naija analogy:** Grading your own exam script with the marking guide on your desk.
- **Example:** Training misses had a spread of 2.84; misses on weeks it never saw had 6.99.

W5D3

📌 Index (DatetimeIndex)

- **Plain English:** The row labels of a table. When they're dates, pandas can do time tricks with them.
- **Naija analogy:** The page numbers of a book. Put dates there and you can jump straight to "August 2024."
- **Example:** `index_col="timestamp"` makes the dates the index.

W1D1

💯 Index to a baseline

- **Plain English:** Divide every value by a starting value and multiply by 100, so everything starts at 100 and you see *change*.
- **Naija analogy:** Comparing students by how much they improved since last term, not by raw score.
- **Example:** `wide / wide.iloc[0] * 100` : South East fell furthest from its own start.

W1D2

🔢 Intercept

- **Plain English:** The starting number a regression adds before the weighted columns. It often has no meaning on its own.
- **Naija analogy:** The "call-out fee" a mechanic charges before he even touches the car.
- **Example:** −22.1 in today's model. Never read it alone.

W5D3

🧹 Interpolation (filling a gap)

- **Plain English:** Filling a missing value with something between its neighbours, usually the straight-line average. It's an assumption, so you must say you did it.
- **Naija analogy:** You missed Tuesday's class, so you copy the notes from Monday and Wednesday and guess what came in between.
- **Example:** Ada's lost week (10 Mar 2025) became 50 tickets, between 48 and 52. Filling it with 0 would invent a week with no faults.

W6D3

🔎 Investigation

- **Plain English:** Answering one question with data, from loading the file to writing the conclusion.
- **Naija analogy:** A mechanic's full diagnosis, not just "e don spoil".
- **Example:** "Is our battery getting tired?": load → profile → plot → smooth → decompose → ACF → finding.

W2D3

🌲 Isolation forest

- **Plain English:** A detector that draws random cuts through the data and counts how many cuts it takes to get each point alone. **Few cuts = odd.** 200 random trees vote.
- **Naija analogy:** Spot the man in agbada at a beach party: one question ("who is wearing agbada?") and he's alone. A guest in shorts takes many questions.
- **Example:** In our demo the agbada point was isolated in 1 cut; a guest in the crowd took 9.

W4D2

📍 k-means

- **Plain English:** A clustering method: choose k centres, put each row with its nearest centre, move the centres, repeat.
- **Naija analogy:** Four bus stops in a new estate: people walk to the nearest one, then the stops are moved to where the crowds are.
- **Example:** `KMeans(4, n_init=10, random_state=42)`.

W4D2

📶 Ka-band

- **Plain English:** A high-frequency radio band that carries lots of data but is very sensitive to rain.
- **Naija analogy:** A sports car: very fast, but it hates bad roads (here: heavy rain).
- **Example:** The Ka-band Nigeria spot beam in today's dataset.

W1D1

🧺 Kitchen-sink feature set

- **Plain English:** Throwing every column you can think of at the model and hoping it sorts them out.
- **Naija analogy:** Adding every spice in the market to one pot of stew. More is not tastier.
- **Example:** The code book's 32 columns caught 3 of 5 events and missed the two easiest; our 5 caught all 5.

W4D3

🥚 Lag

- **Plain English:** How many steps back you look. Lag 1 = one step ago, lag 7 = seven steps ago.
- **Naija analogy:** "Last week this time" when people gist about what happened last week at this same hour.
- **Example:** On daily data, lag 7 = the same weekday last week. On hourly data, lag 24 = the same hour yesterday.

W2D2

⏰ Lag (delay)

- **Plain English:** How late a smoothed line reacts to a real change. For a trailing window, about half the window.
- **Naija analogy:** News reaching the village two weeks after it happened in Abuja.
- **Example:** The trailing 30-day peak came 14 days after the centred one.

W1D3

🚰 Leaking the answer

- **Plain English:** Giving a detector a column that is a reaction to the thing you're trying to detect, so it cheats without knowing.
- **Naija analogy:** Judging who studied for the exam by who looks tired the next morning. You're measuring the after-effect, not the cause.
- **Example:** `uplink_power_dbm` rises because it rains (correlation 0.97). Feed it in and you build a rain detector.

W3D1

🎯 Lean features

- **Plain English:** A small set of columns, each answering a question an engineer would ask.
- **Naija analogy:** A good doctor asks five sharp questions, not fifty random ones.
- **Example:** `lean_features()` : array current, bus voltage, state of charge, battery temperature, temp_per_drive.

W4D3

🪜 Level shift

- **Plain English:** The series jumps to a new normal and stays there.
- **Naija analogy:** Rent goes up and never comes back down.
- **Example:** The amplifier runs about 2 °C hotter from May 2025 onward.

W3D1 W3D2

⚖️ Like with like

- **Plain English:** Only compare two forecasts on the same weeks, the same horizon and the same information.
- **Naija analogy:** Don't compare a student who wrote WAEC with one who wrote a class test.
- **Example:** "Half of naive's error" (5.85 vs 11.25) was unfair; like with like it's 5.85 vs 6.75. And the best baseline (2.24) beats both.

W5D3

🧭 Limitation

- **Plain English:** An honest statement of where your tool stops working: what it will miss, how late it is (or how far ahead it can see), and what you assumed was normal.
- **Naija analogy:** The mechanic who says "this one go hold till December, but no carry am go Kano." You trust him more, not less.
- **Example:** "Its 90% band is about ±7 tickets a week, and it assumes the rainy-season pattern repeats."

W6D3

⛽ Link margin (dB)

- **Plain English:** How much signal is left above the minimum the receiver needs. Below 0 dB the link is down.
- **Naija analogy:** Fuel above the reserve line. Positive: you can still drive. Zero: you're pushing the car. 🚗
- **Example:** `link_margin_db` in `rf_link_quality_hourly`.

W1D2

🪄 Logarithm (log)

- **Plain English:** A maths function that turns multiplying into adding. `np.log()` in Python.
- **Naija analogy:** A translator who turns "times" talk into "plus" talk so everybody understands.
- **Example:** `np.log(demo["multiplicative"])`: the swing stops growing.

W2D1

📏 Long format

- **Plain English:** One row per group per time: many rows, few columns. How databases usually export.
- **Naija analogy:** A class register written as one line per student per day.
- **Example:** `vsat_uptime_by_region_weekly`: 312 rows = 52 weeks × 6 zones.

W1D2

🍀 Lucky catch

- **Plain English:** The scorer says "caught", but the flag inside the event had nothing to do with the fault.
- **Naija analogy:** Guessing C in WAEC and getting it. The mark is real; the knowledge isn't.
- **Example:** EV04 (an amplifier fault) was "caught" by 21 eclipse-hour flags, all at 21:00.

W4D2 W5D1

🎰 Lucky split

- **Plain English:** One test period on which a method scores well by chance, not by skill.
- **Naija analogy:** The man who won one bet and now calls himself a tipster.
- **Example:** The flat training average (57.9 every week) scored 7.77 here and "beat" four baselines. Week 6 · Day 1 tests it on many periods.

W5D2 W6D1

🤖 Machine learning (ML)

- **Plain English:** Instead of writing the rule yourself ("alarm below 20 A"), you show the computer lots of examples and it works out the pattern.
- **Naija analogy:** Nobody gave the Lagos danfo conductor a rule book. He has watched ten thousand passengers, and he knows who will not pay before they climb in.
- **Example:** The isolation forest learns what a normal hour looks like from January–August, then judges the whole year.

W4D2

📐 MAD (median absolute deviation)

- **Plain English:** The median distance from the median: a wobble measure that ignores a few wild points.
- **Naija analogy:** Asking "how far do people usually live from the office?" without letting the one person who commutes from Ibadan spoil the answer.
- **Example:** Robust σ stayed 0.09 A while ordinary σ jumped to 1.08 A after a dropout.

W3D2

🔪 MAE (mean absolute error)

- **Plain English:** The average size of a forecast's mistakes, ignoring whether they were too high or too low.
- **Naija analogy:** How many minutes late your friend usually is, on average. 😅
- **Example:** `(s - s.shift(7)).abs().mean()`

W2D2 W5D2

🎯 Median

- **Plain English:** The middle value when you sort the numbers. Wild values can't drag it.
- **Naija analogy:** The salary of the middle person in the office. The MD's big salary doesn't move it.
- **Example:** The rolling median of the last 168 sunlit hours.

W3D2

🛠️ Method drafted (Checkpoint 3)

- **Plain English:** The first working version of your project's method, running end to end on your data, with your features, reference period and baseline written down.
- **Naija analogy:** The first coat of paint. Not the finish, but you can see the house.
- **Example:** A detection brief brings a statistical baseline plus a multivariate detector on trailing features.

W4D3

🙈 Miss (false negative)

- **Plain English:** A real event the detector never flagged.
- **Naija analogy:** The guard was sleeping when the thief came.
- **Example:** Every value-based detector missed EV06, the battery fade speeding up.

W4D1

〰️ Moving average (centred)

- **Plain English:** The average of a window of values around each point. Classic decomposition uses one to find the trend, so the trend is missing for half a window at each end.
- **Naija analogy:** Judging a student by the average of the tests around each week. You can't do it for the first and last weeks.
- **Example:** The Ka-band trend is empty for the first and last 6 months.

W2D1

🧱 Moving average forecast

- **Plain English:** The average of the last few values (the window), held flat for the whole horizon.
- **Naija analogy:** Averaging your last 8 weeks of transport money to plan next week.
- **Example:** The mean of the last 8 training weeks: 77 tickets. MAE 14.25.

W5D2

🧱 Moving (rolling) average

- **Plain English:** The average of the last N values, recalculated every day as the window moves forward.
- **Naija analogy:** Your average fuel spend over the last 7 days. One big fill-up on Monday doesn't make you panic.
- **Example:** The 7-day average of NOC tickets sits around 6 a day in the dry season.

W1D3

✖ Multiplicative model

- **Plain English:** observed = trend × seasonal × residual. The seasonal swing grows as the level grows.
- **Naija analogy:** December fare goes up 20%. When fares are higher, 20% is more naira.
- **Example:** The `multiplicative` column in `additive_multiplicative_demo`: swing grows from about 0.29 to 0.54.

W2D1

🔀 Multivariate

- **Plain English:** Looking at several channels together, so you can notice a **combination** that never happens, even if each channel looks fine alone.
- **Naija analogy:** NEPA light is on, the inverter is at 93%, and it's not charging. Each fact is normal; together they are wrong.
- **Example:** EV02: array current normal, state of charge stuck at 93% in full sun.

W4D2

🌳 n_estimators

- **Plain English:** How many random trees the forest grows. More trees = steadier scores, slower fit.
- **Naija analogy:** Asking 200 aunties instead of one. One aunty can be wrong; 200 rarely agree on nonsense.
- **Example:** `n_estimators=200`.

W4D2

👵🏾 Naive forecast

- **Plain English:** Repeat the last value you saw for every future step.
- **Naija analogy:** Grandma: "Tomorrow go be like today."
- **Example:** 74 tickets for all 8 weeks (the count on 27 Oct 2025). MAE 11.25.

W5D2

🏭 Named process

- **Plain English:** A specific, real task your unit does regularly, with an owner. Not a general topic.
- **Naija analogy:** Not “food”: “Mama's Saturday jollof for the street party”.
- **Example:** “The weekly battery-health review on the bus subsystem.”

W2D3

🧠 NameError

- **Plain English:** Python's way of saying “I don't know this name.” Usually you skipped a cell, restarted, or forgot quotes.
- **Naija analogy:** Calling someone “Chief” at a party when nobody has introduced him to you yet.
- **Example:** `NameError: name 'ka' is not defined` → run the cells above.

W1D1

🕳️ NaN

- **Plain English:** “Not a Number”. pandas' symbol for an empty or missing value.
- **Naija analogy:** The empty seat at a wedding where the guest didn't come.
- **Example:** After `.asfreq("D")`, the 3 deleted days show `NaN`.

W1D1

👥 Near-identical neighbours

- **Plain English:** In hourly data, 10:00 and 11:00 look almost the same. Shuffle the hours and every test hour has a near-twin in training.
- **Naija analogy:** Writing an exam beside your twin who has the same answers you studied together.
- **Example:** Quiz 3, Q23: why a random split on hourly telemetry scores too well.

W5D1

🐱 Nearest neighbour ("the copycat")

- **Plain English:** A model that answers by finding the most similar example it has seen and copying its answer. With `n_neighbors=1` it copies just one.
- **Naija analogy:** The student who copies from whoever sits closest.
- **Example:** `KNeighborsClassifier(n_neighbors=1)` recognised EV05 because it had seen EV02, but was blind to EV04, a kind of fault it had never seen.

W5D1

🎧 NOC (Network Operations Centre)

- **Plain English:** The team that watches the network 24/7 and opens a ticket for every fault.
- **Naija analogy:** The control room of a big estate, where security watches all the CCTV.
- **Example:** `noc_tickets_daily.csv` counts their tickets.

W1D3

🎲 Noise

- **Plain English:** Random wobble that means nothing. Don't react to it.
- **Naija analogy:** Weighing yourself after eating amala: 2 kg up doesn't mean you're getting fat. 😄
- **Example:** NOC tickets 38 one week, 41 the next = the same week twice.

W1D1 W5D2

☕ Noise (again)

- **Plain English:** Random week-to-week wobble with no meaning. Don't react to it.
- **Naija analogy:** Your weight after a heavy plate of pounded yam. Wait a week before you worry.
- **Example:** 38 vs 41 tickets a week, when the mean is about 36 and the usual swing is about 5.

W1D3

🚫 None ("no crossing")

- **Plain English:** Python's way of saying "there is no answer". Our `day_of_threshold` returns `None` instead of a fake day 0.
- **Naija analogy:** "No light" is an answer. Pretending NEPA brought light at midnight is not.
- **Example:** `day, line = day_of_threshold(known["propellant_kg"], 5.0)` gives `day = None`.

W6D2

📓 Notebook

- **Plain English:** A document that mixes reading (text) and code you can run, in one page.
- **Naija analogy:** A recipe book where you can cook each step right on the page and taste it immediately.
- **Example:** Files ending in `.ipynb` are notebooks.

W1D1

🐞 np.argmax on True/False

- **Plain English:** `np.argmax` returns the position of the largest value. On True/False it finds the **first True**. If there is **no** True, it quietly returns **0**.
- **Naija analogy:** Asking the gateman "who was the first visitor?" on a day nobody came, and he answers "the first name on the list" anyway.
- **Example:** The original code book printed 1 Jan 2021 (day 0) as the propellant's crossing date.

W6D2

🧱 np.polyfit (slope and intercept)

- **Plain English:** `np.polyfit(x, y, 1)` finds the best straight line through the points and returns two numbers: the **slope** (change per step) and the **intercept** (the value at step 0).
- **Naija analogy:** Slope = how much the price of a bag of rice goes up each month; intercept = what it cost when you started counting.
- **Example:** Battery slope −0.0126% per day, which is −4.59% a year; intercept 99.88%.

W6D2

🏁 Onset (start date)

- **Plain English:** The day a change really began, as opposed to the day the alarm rang.
- **Naija analogy:** The day your tyre started leaking vs the day you noticed it was flat.
- **Example:** The practice fault began 10 Feb 2025; the alarm rang 13 Feb 2025.

W3D3

🌗 Operating state

- **Plain English:** A distinct mode the system runs in, each with its own normal behaviour.
- **Naija analogy:** Your house on NEPA light vs on generator vs on inverter: different "normal" noise, different normal bills.
- **Example:** eclipse (~0 A), recharge (~29.6 A), sunlit (~28.1 A).

W3D2

🚧 Operational limit (threshold)

- **Plain English:** The agreed value a reading must not cross, set by engineering, known up front. You forecast *when* it's reached, not what it is.
- **Naija analogy:** The red line on the fuel gauge: nobody forecasts where the red line is.
- **Example:** `degradation_thresholds.csv`: battery 80%, amplifier −2 dB, propellant 5 kg.

W6D2

❌ Outage

- **Plain English:** The link is actually down: link margin below 0 dB.
- **Naija analogy:** Not "the network is slow", but "no network at all".
- **Example:** 12 outage hours in August, each lasting one hour.

W3D1

🐼 pandas

- **Plain English:** The Python library for working with tables of data.
- **Naija analogy:** Microsoft Excel, but you control it by typing instructions, so it never gets tired.
- **Example:** `import pandas as pd`, then `pd.read_csv(...)`.

W1D1

🗓️ parse_dates

- **Plain English:** Tells pandas “this column is a date, not just text.”
- **Naija analogy:** Telling the gateman, “this is not just any paper, it's my ID card.”
- **Example:** `pd.read_csv(..., parse_dates=["timestamp"])`

W1D1

✂️ Partial period

- **Plain English:** A bucket at the edge of the data that doesn't have all its days.
- **Naija analogy:** Judging a football match at half-time and saying the team scored few goals.
- **Example:** The last weekly bucket has only 2 days, so it shows just 9 tickets.

W1D3

🍹 Period

- **Plain English:** How many rows until the seasonal pattern repeats. It comes from the process, not from a default.
- **Naija analogy:** How often your esusu (contribution) group meets: weekly means 7 days, monthly means 12 times a year.
- **Example:** 24 for hourly bus power, 7 for daily tickets, 12 for monthly availability.

W2D1 W2D2

📐 Period (season length)

- **Plain English:** How many steps until the pattern repeats.
- **Naija analogy:** Harmattan comes back every 12 months, not every 7 days.
- **Example:** Weekly data with a yearly season: `period = 52`. There's no weekday pattern in a weekly series.

W5D2

🧩 Persistence (memory)

- **Plain English:** A series "has memory" when its past values help you guess its next values.
- **Naija analogy:** A neighbour who has been borrowing sugar every week will likely knock again next week. 😅
- **Example:** Throughput has strong weekly memory; random noise has none.

W2D2

⏳ Persistence rule

- **Plain English:** Only alarm if the reading stays bad for N readings in a row.
- **Naija analogy:** Not shouting "NEPA has taken light!" until the light has been off for a full minute.
- **Example:** A 2-hours-in-a-row rule caught 0 of the 12 one-hour outages in August.

W3D1

🔗 Pipeline

- **Plain English:** Gluing the scaler and the model together, so the scaler fitted on the training rows is exactly the one used at prediction time.
- **Naija analogy:** The tailor keeps your measurements with your cloth. Nobody sews your agbada with someone else's tape-measure numbers.
- **Example:** `make_pipeline(StandardScaler(), KMeans(4, n_init=10, random_state=42))`.

W4D2

🔄 pivot

- **Plain English:** The pandas command that reshapes long → wide.
- **Naija analogy:** Rearranging wedding guests from one long queue into tables by family.
- **Example:** `vsat.pivot(index="week_start", columns="region", values="uptime_pct")`

W1D2

🎯 Planning figure

- **Plain English:** A forecast stated as a range, with how far it reaches and when it will be re-checked. Built for decisions, not for showing off precision.
- **Naija analogy:** "We'll travel for Christmas sometime in mid-December, I'll confirm by November."
- **Example:** "Amplifier: around 2033 (calls span March to November), more than 7 years past our data; we re-estimate every year."

W6D2

📐 plot_acf / statsmodels

- **Plain English:** `statsmodels` is a Python library for statistics; `plot_acf` is its correlogram function. Colab already has it.
- **Naija analogy:** A specialist mechanic: pandas is the general workshop, statsmodels is the one who knows engines.
- **Example:** `from statsmodels.graphics.tsaplots import plot_acf`

W2D2

〰️ Polynomial fit (degree 2)

- **Plain English:** A curve with one bend instead of a straight line: `np.polyfit(x, y, 2)`. It can follow a slowing fade, but it can go wild when stretched far into the future.
- **Naija analogy:** A road that bends with the hill. Great on this hill; don't assume it keeps bending the same way to Abuja.
- **Example:** On the same three years, a degree-2 curve calls the battery crossing on 14 Jun 2025, inside the real window.

W6D2

🧪 Practice (injected) fault

- **Plain English:** A small fault you add to clean data on purpose, so you know the right answer and can test your detector.
- **Naija analogy:** A fire drill: you know there is no fire, but you learn exactly how fast everyone gets out.
- **Example:** `practice_residuals()` hides +0.3 °C in the amplifier's calm leftovers.

W3D3

📟 Precision

- **Plain English:** Of all the alarms, the fraction that were real. "When it pages me, is it real?"
- **Naija analogy:** Of all the times Musa shouted "Thief!", how many were actual thieves?
- **Example:** Musa: 9 ÷ 200 = 0.045. All-hours control chart: 0.06.

W4D1

🔮 predict

- **Plain English:** Asking the trained model about rows. For an isolation forest: **-1 means odd**, 1 means looks normal.
- **Naija analogy:** After his month of watching, the apprentice says "this engine sound no correct".
- **Example:** `model.predict(X) == -1` gives the True/False flags.

W4D2

↔️ Prediction interval (90% band)

- **Plain English:** A range around a forecast that the real value should land inside a stated share of the time. A 90% band should be right about 9 times in 10.
- **Naija analogy:** "30 to 90 minutes, Third Mainland permitting" instead of "I'm on my way".
- **Example:** The rate forecast for the week of 3 Nov: 56, band 49 to 63. Actual: 53.

W5D3

🚨 Priority-1 ticket

- **Plain English:** The most urgent kind of fault ticket. Rare, but serious.
- **Naija analogy:** A call from your MD at 11pm: it doesn't happen often, but when it does, everybody moves.
- **Example:** About 1 a day, and 17% of days have none.

W1D3

👨🏾‍💼 Process owner

- **Plain English:** The person responsible for that process, who will use or approve your tool.
- **Naija analogy:** The oga at the top of that particular desk.
- **Example:** The shift lead who runs NOC fault escalation.

W2D3

🔍 Profiling

- **Plain English:** Quick checks before you trust a dataset: rows, date range, gaps, duplicates, missing values.
- **Naija analogy:** Inspecting a tokunbo car before you pay: engine, tyres, papers.
- **Example:** `len(ka)`, `ka.index.min()`, `ka.isna().sum()`.

W1D1

📊 Quantile (90th percentile)

- **Plain English:** The value that a given share of the numbers falls below. The 90th percentile of the misses: 90% of misses were smaller than this.
- **Naija analogy:** In a class of 10, the score that only the top student beats.
- **Example:** `np.quantile(np.abs(misses), 0.9)` = 8.20, so the honest band is forecast ± 8.20, and it covered 7 of 8.

W5D3

🌧️ Rain fade

- **Plain English:** Heavy rain absorbs the satellite signal, so quality drops or the link goes down.
- **Naija analogy:** DStv showing "No signal" during a heavy Lagos downpour.
- **Example:** The yearly dips in availability, worst in July and August.

W1D1 W1D2

✂️ Random cut

- **Plain English:** One split of the data along one column, at a random value between that column's smallest and largest value.
- **Naija analogy:** "Everyone taller than this, step left." Repeat with random questions until one person is alone.
- **Example:** One cut on array current at 10 A puts all 92 eclipse hours (at most 0.05 A) on one side and every other hour (at least 16 A) on the other.

W4D2

🎲 Random split

- **Plain English:** Picking the test rows at random, for example with `train_test_split`. Fine for unrelated rows, wrong for time series.
- **Naija analogy:** Mixing the exam questions into the practice pile and hoping nobody notices.
- **Example:** `train_test_split(X, y, test_size=0.33, random_state=42)` put hours of EV04 on both sides.

W5D1

🚫 Random-split leakage

- **Plain English:** If you split time data randomly, the test set's near-copies sit in the training set, so the answers leak.
- **Naija analogy:** WAEC expo: if you saw the questions before the exam, your score is fake.
- **Example:** Day 8 in test, days 1 and 15 (same weekday, r = 0.99) in training.

W2D2

🎲 random_state

- **Plain English:** A fixed starting number for the randomness, so the model gives the same answer every run.
- **Naija analogy:** Shuffling the Ludo dice the same way every time, so everyone gets the same game.
- **Example:** `random_state=42`. With 0 or 1 instead, the steady-sun forest loses its one EV04 hour.

W4D2

🦏 Rare events (class imbalance)

- **Plain English:** When the thing you're hunting is tiny compared with everything else, so "always say no" looks great.
- **Naija analogy:** Looking for 6 fake ₦1,000 notes in a bag of 4,706 genuine ones.
- **Example:** 6 events in 4,706 readings.

W4D1

🛟 Rare is not broken

- **Plain English:** A model finds what is unusual. Some unusual things are normal physics with a known cause.
- **Naija analogy:** The lifeguard in red shorts stands out at the beach every evening. He belongs there.
- **Example:** The raw forest's 58 flags were all 21:00 eclipse hours.

W4D2

➗ Rate (per-size number)

- **Plain English:** A count divided by the size of what produces it. Rates repeat when counts don't, because the growth is divided out.
- **Naija analogy:** NEPA bill **per room**, fuel **per kilometre**, jollof **per guest**.
- **Example:** Tickets per 1,000 terminals: January 25.6 in 2024, 25.9 in 2025.

W5D2

🎯 Rate seasonal naive

- **Plain English:** Seasonal naive on the rate: last year's tickets per terminal × today's terminals.
- **Naija analogy:** Grandma, adjusted for the bigger family: "Last Christmas each guest ate two plates. This year we have 35% more guests."
- **Example:** 42 tickets ÷ 1,588 terminals × 2,131 terminals = 56.4 for the week of 3 Nov (actual 53). MAE 2.24 over the 8 weeks.

W5D2

📦 Raw data

- **Plain English:** The original export from a system, before you clean or summarise it.
- **Naija analogy:** Unpeeled yam straight from the market.
- **Example:** Never commit raw NIGCOMSAT exports to GitHub: commit only your code.

W2D3

📘 README and requirements.txt

- **Plain English:** Two files every project repository needs. `README.md` says what the tool does and how to run it; `requirements.txt` lists the Python packages it needs.
- **Naija analogy:** The manual and the parts list that come with the new generator. Without them, only the person who installed it can start it.
- **Example:** The README states where the tool runs on the training server and its URL or schedule.

W6D3

🕸️ Recall

- **Plain English:** Of all the real events, the fraction the detector caught. "Did we miss anything?"
- **Naija analogy:** Of the 10 thieves who came this year, how many did the guard catch?
- **Example:** Musa: 9 ÷ 10 = 0.90. Sunlit control chart: 3 ÷ 3 = 1.00.

W4D1

⚖️ Recall-first vs precision-first

- **Plain English:** Deciding which number matters more for a system, based on what a miss costs vs what a false alarm costs.
- **Naija analogy:** A smoke alarm should be recall-first (better a false alarm than a burnt house). A car alarm that screams at every keke is a precision problem.
- **Example:** Battery and power: recall first. Rain-fade dashboards: precision first.

W4D1

⚡ Recharge hour

- **Plain English:** The hour after an eclipse, when the panels push extra current to refill the battery and the battery warms up.
- **Naija analogy:** When NEPA brings light back and every phone and power bank in the house charges at once.
- **Example:** Battery temperature goes above 22 °C in the hour after every eclipse: normal.

W3D1

🔋 Recharge hours

- **Plain English:** The hours just after an eclipse when the solar array works extra to refill the battery.
- **Naija analogy:** Charging your phone to 100% the moment light comes back.
- **Example:** 138 recharge hours in the bus-power file.

W3D2

🏆 Recommend the simpler model

- **Plain English:** If a simple method wins or ties a complex one on a fair test, you recommend the simple one: easier to explain, check and maintain.
- **Naija analogy:** If the keke gets you there as fast as the SUV, take the keke.
- **Example:** The 3-sigma chart on the lean features: both test events, 1 false alarm. The forest: both events, 25 false alarms.

W5D1

🗣️ Recommendation (the 4 sentences)

- **Plain English:** The ending of your report and demo: **S**ituation, **O**bservation (with one number), **I**mplication, **R**ecommendation (what to do, with which tool).
- **Naija analogy:** The uncle in the front row: "Tell us in one sentence: what does it do, and why should we care?" Four sentences is your allowance.
- **Example:** "Use the rate forecast for the monthly roster, keep 'same as last week' for next week, and re-run the backtest every quarter."

W6D3

📅 Reference period

- **Plain English:** A calm, known-good stretch of data used to learn what normal looks like.
- **Naija analogy:** Judging your new neighbour by a quiet fortnight, not by the night of their wedding party.
- **Example:** 1–15 August 2025: sunlit, no eclipse yet, nothing strange.

W3D2 W5D1

🔁 Refit

- **Plain English:** Training the model again from scratch at each fold, on only the data before that fold's cut.
- **Naija analogy:** A coach who picks the team fresh each week from the latest form, not from last season's line-up.
- **Example:** The code book refits every 4 weeks. Refitting every week instead moves the regression from losing to naive (5 of 6 folds) to a near tie (2.96 vs 3.16).

W6D1

📈 Regression forecast

- **Plain English:** A regression that predicts next week's value from columns that are all known before the week starts.
- **Naija analogy:** A danfo conductor guessing today's passengers from yesterday's crowd, last December's crowd and how many new estates opened on the route.
- **Example:** `LinearRegression` on lag1, lag52, terminals, sin52 and cos52 forecast the 8 held-out weeks with MAE 5.85.

W5D3

📈 Regression (line of best fit)

- **Plain English:** A way to learn a recipe like "temp ≈ base + k × drive" from past data.
- **Naija analogy:** Working out from past trips that every extra 10 km adds about the same to your fuel cost.
- **Example:** temp ≈ 49.5 + 0.17 × drive + sun-angle wave.

W3D2

🔗 Relationship feature (cross-channel feature)

- **Plain English:** One column built from two or more sensors that normally move together, so the model sees when they stop agreeing.
- **Naija analogy:** NEPA has brought light but the inverter is still at 93%. Light alone is fine; 93% alone is fine (it happens every night). Together, in daylight, the charger is faulty.
- **Example:** `temp_per_drive`, and the steady-sun state for state of charge.

W4D3

⛽ Remaining useful life (RUL)

- **Plain English:** How long a part has left before its reading crosses an operational limit. Usually: (what's left above the limit) ÷ (how fast it's going).
- **Naija analogy:** The danfo driver: 20 litres left, 4 litres a trip, so 5 more trips before he's pushing the bus.
- **Example:** On 1 Jan 2024, the battery's line reaches 80% on 1 May 2025: a remaining useful life of 486 days.

W6D2

📦 Repository (repo)

- **Plain English:** A project folder tracked by Git, often stored on GitHub, holding your code and its history.
- **Naija analogy:** Your project file at the registry, with every version stamped and dated.
- **Example:** Your final project's code, `requirements.txt` and `README.md` live in a repo.

W2D3

🔁 resample

- **Plain English:** Change the time step of a series (hourly → daily) and choose how to summarise each new step.
- **Naija analogy:** Turning a daily sales diary into a weekly report: you must decide to add, average or take the best day.
- **Example:** `.resample("D").min()`

W1D2 W1D3

🔄 Reset

- **Plain English:** Setting the running total back to zero after someone has checked an alarm.
- **Naija analogy:** Clearing the tab at the mama put after you've paid. You don't keep paying for last week.
- **Example:** Without a reset, CUSUM alarms every single day after the real step.

W3D3

🎲 Residual

- **Plain English:** What's left after the trend and season are removed. Mostly noise, but big residuals are surprises worth investigating.
- **Naija analogy:** After you've explained the traffic by "it's Monday" and "it's raining", whatever delay is left is the keke that broke down.
- **Example:** In August, the biggest residual points straight at an injected solar-array dropout.

W2D1

↩️ Response (vs cause)

- **Plain English:** Something that changes *because of* something else.
- **Naija analogy:** You carry an umbrella because it's raining. The umbrella didn't bring the rain.
- **Example:** Uplink power is a response to rain.

W1D2

🔒 Restricted data

- **Plain English:** Real NIGCOMSAT data that must not leave NIGCOMSAT systems or go into outside tools without permission.
- **Naija analogy:** Your ATM PIN: useful, but you don't paste it into WhatsApp.
- **Example:** Real telemetry, RF records and network logs. Course data is synthetic and safe.

W2D3

👩🏾‍🔬 Right reason

- **Plain English:** A flag is only useful if the feature that triggered it matches the real cause. Check with z-scores.
- **Naija analogy:** A student who gets the right answer by copying hasn't learnt anything. Show your working.
- **Example:** Every EV04 hour the forest flagged was flagged by `temp_per_drive`, exactly as the physics says.

W4D3

🛡️ Robust

- **Plain English:** A method that doesn't let a few extreme values pull the result around.
- **Naija analogy:** One bad experience at one buka shouldn't make you blacklist every buka in Lagos.
- **Example:** With one fake disaster August, robust STL keeps August's seasonal effect near −1.2 instead of −3.0.

W2D1

🧭 Robust z-score

- **Plain English:** How many robust σ a reading sits from the rolling median.
- **Naija analogy:** "How many times more than usual is this bill?"
- **Example:** The dropouts sat about 140 robust σ below normal.

W3D2

📊 Rolling-mean crossing

- **Plain English:** The first day the average of the last N days (here 30) is below the limit. Slower, but noise can't fool it.
- **Naija analogy:** Judging a footballer by his last 30 matches, not his worst one.
- **Example:** 20 Jun 2025. The 3-year call is 50 days early against it.

W6D2

🔁 Rolling origin (TimeSeriesSplit)

- **Plain English:** Testing several times: each fold trains on everything before a block and tests on that block. Training only ever grows **forwards**. You get a spread of scores, not one lucky number.
- **Naija analogy:** Judging a striker on a whole season, not on one match.
- **Example:** `TimeSeriesSplit(n_splits=4, gap=24)` : the chart's false alarms per fold were 1, 23, 1 and 1.

W5D1

🏎️ Rolling slope

- **Plain English:** The speed of change over the last N days, recalculated every day.
- **Naija analogy:** Your car's speedometer: it tells you how fast you're going right now, not how far you've gone.
- **Example:** `rolling_slope(cap, 90)` crossed −3.0 Ah/yr on 2 May 2025.

W3D3

〽️ Rolling standard deviation

- **Plain English:** How much the values jumped around inside the window. A measure of wobble.
- **Naija analogy:** How much your neighbour's generator noise goes up and down in one evening.
- **Example:** `s.rolling(7).std()`

W1D3

🌊 Rolling window

- **Plain English:** A fixed number of rows that slides along the series one row at a time. You calculate something (usually the average) inside it.
- **Naija analogy:** A danfo with exactly 7 seats: every time one passenger enters at the front, the oldest one drops at the back.
- **Example:** `tk["tickets_raised"].rolling(7).mean()`

W1D3

🕳️ Rows lost to the window

- **Plain English:** A 24-hour window or a 24-hour lag has nothing to work with during the first day, so those rows are empty (NaN) and get dropped.
- **Naija analogy:** You can't give a weekly average on Monday morning.
- **Example:** The 32-column builder turns 8,760 rows into 8,736.

W4D3

📋 Rubric

- **Plain English:** The marking scheme: the list of what earns points and how many. Ours has six lines adding up to 100.
- **Naija analogy:** The WAEC marking scheme: the examiner isn't guessing, every mark has a reason.
- **Example:** Honest evaluation is the biggest line: 25 of 100 points.

W6D3

🪜 Run rules

- **Plain English:** Extra alarm rules that look at patterns, not just single points. Rule 1: one point beyond 3σ. Rule 2: nine in a row on one side of the centre.
- **Naija analogy:** One late payment is chance. Nine late payments in a row means something has changed.
- **Example:** Rule 2 is best at catching level shifts.

W3D2

➕ Running total

- **Plain English:** A number you keep adding to, day after day, instead of looking at each day alone.
- **Naija analogy:** Your esusu/ajo contribution book: one day's ₦2,000 is small, the month's total is not.
- **Example:** CUSUM's `S` is a running total of leftovers minus slack.

W3D3

🔄 Runtime / session

- **Plain English:** The live connection to Google's computer, plus everything it currently remembers.
- **Naija analogy:** A phone call. Hang up (restart) and the other person forgets what you were saying.
- **Example:** *Runtime → Restart session* wipes the memory. Run STEP 0 again after it.

W1D1

✅ Sanity check (the count)

- **Plain English:** Before reading results, check the number of flags is roughly what you asked for. Thousands = a bug.
- **Naija analogy:** You sent ₦5,000 and the bank app says ₦500,000 left your account. Don't celebrate or panic: check.
- **Example:** Writing `predict() == 1` by mistake flags 8,702 hours: the normal ones.

W4D2

💾 Save a copy in Drive

- **Plain English:** Makes your own editable copy of the notebook in your Google Drive.
- **Naija analogy:** Photocopying the class notes before writing on them, so you don't scribble on the original.
- **Example:** *File → Save a copy in Drive*. Your copy lands in Drive → *Colab Notebooks*.

W1D1

🔪 Scaling (StandardScaler)

- **Plain English:** Turning every column into "standard deviations from its mean", so no column wins just because its numbers are big.
- **Naija analogy:** Comparing a ₦ price and a $ price without converting first: the naira one always looks bigger.
- **Example:** Without scaling, k-means smears the recharge hours over 3 clusters. The forest doesn't care either way.

W4D2

🗓️ Scheduled report

- **Plain English:** A script the server runs on a timetable, which saves a dated chart and a short summary to a shared folder.
- **Naija analogy:** The newspaper vendor who drops the paper at your gate every morning at 7, whether you remember to ask or not.
- **Example:** `ticket-report-2025-12-22.png` and `ticket-report-2025-12-22.txt` : "about 65 tickets, plausibly 58 to 71".

W6D3

🧰 scikit-learn (sklearn)

- **Plain English:** The free Python library of ready-made machine-learning models. Colab already has it.
- **Naija analogy:** Alaba market for models: whatever you need, someone has a stall, and every stall uses the same POS machine.
- **Example:** `from sklearn.ensemble import IsolationForest`

W4D2

🧑🏾‍⚖️ Scoring rules

- **Plain English:** The agreement on what counts as a catch and what counts as a false alarm, written down **before** anyone computes a score.
- **Naija analogy:** Agreeing the rules of the five-a-side match before kick-off, not after your team concedes.
- **Example:** One flag inside an event = caught; many flags inside one event = one catch.

W4D1

🔎 Search window

- **Plain English:** How far into the future the code looks for a crossing. If the trend crosses after the window ends, the honest answer is "no crossing inside the window".
- **Naija analogy:** Looking for your friend's car only on your own street: not finding it doesn't mean it doesn't exist.
- **Example:** The 4,000-day search ended on 13 Dec 2034; the propellant line reaches 5 kg on 6 Dec 2035.

W6D2

🔁 Seasonal component

- **Plain English:** The pattern that repeats exactly every period: the same 12 numbers every year, or the same 7 numbers every week.
- **Naija analogy:** Rush hour at 7am every weekday, like clockwork.
- **Example:** August is about 1.1 points below the yearly average every year.

W2D1

🔧 seasonal_decompose

- **Plain English:** The classic decomposition function. Simple and fast, but it assumes the season looks the same every cycle.
- **Naija analogy:** A tailor who uses one measurement for you every year, even if you've added weight. 😅
- **Example:** `parts = seasonal_decompose(series, model="additive", period=7)`

W2D1

📆 Seasonal naive

- **Plain English:** Copy the value from exactly one season ago: for weekly data with a yearly pattern, 52 weeks back.
- **Naija analogy:** "Last December the market was full, so this December will be full too."
- **Example:** It copied 42–54 tickets from Nov–Dec 2024, when the network was 35% smaller. MAE 15.25, last place.

W5D2

🗓️ Seasonal naive ("same weekday last week")

- **Plain English:** Forecast each day with the value from one period ago.
- **Naija analogy:** Expecting this Friday's traffic to be like last Friday's traffic.
- **Example:** For throughput its average error is 0.062 Gbps, about 5× better than "same as yesterday" (0.338).

W2D2

🌧️ Seasonality

- **Plain English:** A pattern that repeats on a **fixed calendar**: every day, every week, every year.
- **Naija analogy:** Harmattan every December, rainy season every year, traffic every weekday morning.
- **Example:** Ka-band availability dips every August.

W1D1

🌧️ Seasonality in steps

- **Plain English:** A pattern that repeats every year but switches on and off suddenly, instead of rising and falling smoothly.
- **Naija analogy:** The rainy season doesn't creep in: one week it's dry, the next week the gutters are full.
- **Example:** Tickets jumped +17 in the first week of April 2024 and +16 in April 2025, and dropped −15 in the first week of November 2024.

W5D2

🧼 Series hygiene

- **Plain English:** The basic health check of a time series before any analysis: a time index, sorted, no repeated timestamps, no holes (or holes explained), a declared frequency, and a known time zone.
- **Naija analogy:** Washing your hands before you cook: nobody claps for it, but skip it and everybody gets sick.
- **Example:** Ada's export had one week exported twice and one week that never arrived.

W6D3

📈 Shared trend

- **Plain English:** Two series both rising (or falling) over time for their own reasons, which makes them look linked.
- **Naija analogy:** Your salary and the price of garri both rose over 5 years. One didn't cause the other.
- **Example:** Amplifier temperature and RF drive both drift up over two years.

W2D2

⏪ .shift()

- **Plain English:** The pandas command that makes a lagged copy by sliding every value down.
- **Naija analogy:** Everyone in the queue moving one seat back so a latecomer can sit in front.
- **Example:** `s.shift(1)` puts yesterday's value on today's row.

W2D2

⌨️ Shift + Enter

- **Plain English:** The keyboard shortcut that runs the current cell and moves to the next.
- **Naija analogy:** "Next please!" at the bank queue.
- **Example:** Click a cell, press Shift + Enter, see the output.

W1D1

⏱️ Shift hours (the cost of false alarms)

- **Plain English:** False alarms × minutes to check each one, turned into engineer hours. The number a manager can decide on.
- **Naija analogy:** Counting how much fuel the generator wasted while nobody was home.
- **Example:** 46 per season × 2 seasons × 20 minutes ≈ 31 hours a year, about 4 shifts.

W4D1

🤫 Silent degradation

- **Plain English:** Something wearing out without any alarm firing.
- **Naija analogy:** Tyres going bald slowly. No warning light, until the blow-out.
- **Example:** A 95 Ah capacity limit fires 0 times in three years of fading.

W3D1

📜 SLA (Service Level Agreement)

- **Plain English:** The promise to a customer, e.g. "the link will be up 99.5% of the time".
- **Naija analogy:** The landlord's promise of "24 hours light and water" in the house advert. 😅
- **Example:** "Did we breach the SLA?" is answered with daily **minimums**, not averages.

W1D2 W1D3

🪫 Slack (k)

- **Plain English:** A small allowance taken away every day, so ordinary noise doesn't pile up.
- **Naija analogy:** The "abeg, small change" the bus conductor never returns. Too small to fight over.
- **Example:** `k = 0.5 × σ` = 0.10 °C for the amplifier.

W3D3

📐 Slope alarm (fade-rate alarm)

- **Plain English:** An alarm on how **fast** something is changing, not on its value. It catches drifts that never cross a fixed line.
- **Naija analogy:** Your fuel gauge says half a tank, but it was full an hour ago. The level is fine; the speed of dropping is the problem.
- **Example:** Fade faster than −3 Ah a year → alarm. It catches EV06 with zero false alarms.

W4D1

🦕 Slow decay

- **Plain English:** Bars that start high and fade gently, lag after lag. It's the fingerprint of a trend.
- **Naija analogy:** Lagos rent: this year looks like last year, which looked like the year before… all going one way.
- **Example:** Transponder utilisation: 0.90, 0.81, 0.73, 0.64…

W2D2

🦕 Slow drift

- **Plain English:** A gradual change, a little every day. No single day looks unusual.
- **Naija analogy:** Your trousers getting tighter small-small since December. 😅
- **Example:** A rolling "expected" follows a slow drift, so its residual never notices.

W3D2

🖼️ Small multiples

- **Plain English:** A grid of small, identical charts, one per group, sharing the same y-axis.
- **Naija analogy:** Passport photos of the whole family: same size, same background, so you can compare faces.
- **Example:** Six mini charts, one per geopolitical zone, `sharey=True`.

W1D2

🌀 Smoothing

- **Plain English:** Removing the small, fast wiggles so the slower, real movement is easier to see.
- **Naija analogy:** Ironing a shirt: the shape is still there, the creases are gone.
- **Example:** The 7-day line shows the rainy-season rise hidden inside the daily zig-zag.

W1D3

🌞 Solar-array current

- **Plain English:** The electric current coming from the satellite's solar panels. About 27.8 A in full sunlight.
- **Naija analogy:** The output of your rooftop solar panel on a sunny afternoon.
- **Example:** A sudden drop to about 16 A with no eclipse is a fault: a "string dropout".

W2D1

🍝 Spaghetti chart

- **Plain English:** Too many lines on one chart. Nobody can follow any of them.
- **Naija analogy:** Ten danfo routes drawn on one map with the same blue pen.
- **Example:** Six zones on one axis: try finding South South.

W1D2

🔢 Sparse (small) counts

- **Plain English:** Counts that are usually 0, 1 or 2. One extra event is a big jump, so they need wider windows.
- **Naija analogy:** Counting how many times NEPA brings light at 3am. Mostly zero, sometimes one.
- **Example:** Priority-1 tickets per day.

W1D3

🚨 Special-cause variation

- **Plain English:** A change with a real reason behind it: worth investigating.
- **Naija analogy:** Traffic not moving at all because a tanker fell on the bridge.
- **Example:** The array dropping to 15.3 A in full sun.

W3D2

📈 Spike

- **Plain English:** A bar much taller than its neighbours, sticking above the shaded band.
- **Naija analogy:** The one day in the month when everyone rushes to the bank: salary day.
- **Example:** Throughput spikes at lags 7, 14, 21, 28, 35.

W2D2

⏳ Split by time (temporal split)

- **Plain English:** Every training hour comes **before** every test hour. It's the only honest split for time series, because tomorrow is always unseen.
- **Naija analogy:** You study the past, then you face tomorrow's exam. You can't study tomorrow.
- **Example:** `train = bus.index < "2025-09-01"`, `test = bus.index >= "2025-09-02"`.

W5D1

🔦 Spot beam

- **Plain English:** A satellite beam aimed at one region, not the whole continent.
- **Naija analogy:** A torchlight pointed at one spot instead of a bulb lighting the whole room.
- **Example:** “Nigeria spot beam” in the `beam` column.

W1D1

📊 Spread across folds

- **Plain English:** The range of scores from best fold to worst fold, and when the bad ones happened. It tells the user how much to trust the forecast, and when not to.
- **Naija analogy:** A striker who "averages a goal a game" but scored five in one match and none in the other nine.
- **Example:** The regression's fold MAEs run from 1.27 to 7.67 (mean 4.01); the worst fold contains 3 November, when tickets fell from 74 to 53.

W6D1

👹 Spurious correlation

- **Plain English:** A strong correlation between two things that are NOT really linked.
- **Naija analogy:** Suya sales and generator sales both rise toward December. Suya doesn't sell generators. 😄
- **Example:** Made-up suya vs generators: r = 0.97.

W2D2

🥖 Stale reference

- **Plain English:** Limits learned long ago that no longer match today's normal.
- **Naija analogy:** Last year's price list at the market: everything looks "too expensive" now.
- **Example:** 1,861 of 2,696 sunlit hours fall outside the August band by November.

W3D2

📐 Standard deviation (σ, "sigma")

- **Plain English:** The typical size of the wobble around the average.
- **Naija analogy:** How much your danfo fare moves around the usual price on a normal day.
- **Example:** Sunlit array current has σ ≈ 0.59 A when all sunlit hours are counted.

W3D1 W3D2

🔋 State of charge (SoC)

- **Plain English:** How full the battery is, in percent.
- **Naija analogy:** The battery bar on your phone.
- **Example:** `battery_soc_pct` drops to 77.3% on the deepest eclipse night.

W3D1

🚀 Station-keeping burn

- **Plain English:** A short thruster firing that pushes a geostationary satellite back into its assigned box. Each burn uses a little propellant.
- **Naija analogy:** The danfo driver nudging back into his lane every few minutes on Third Mainland Bridge.
- **Example:** About one every 14 days, roughly 0.39 kg each, about 26 a year.

W6D2

📦 statsmodels

- **Plain English:** A Python library for statistics and time-series models. Already installed in Colab.
- **Naija analogy:** The big toolbox in the mechanic's shop. pandas is the spanner; statsmodels has the special tools.
- **Example:** `from statsmodels.tsa.seasonal import seasonal_decompose`

W2D1

🌞 Steady sun

- **Plain English:** Hours with no eclipse in this hour or the 3 before: not on battery, not recharging. The state we judge today.
- **Naija analogy:** Not "NEPA just took light" and not "NEPA just brought light and the inverter is charging": normal daytime.
- **Example:** `bus["eclipse_flag"].rolling(4).max() == 0`: 8,389 of 8,760 hours.

W4D2

🧠 STL

- **Plain English:** A smarter decomposition (Seasonal-Trend decomposition using LOESS) that lets the season change slowly over time.
- **Naija analogy:** A tailor who measures you again before every new outfit.
- **Example:** `STL(series, period=24, robust=True).fit()`

W2D1

📊 Streamlit dashboard

- **Plain English:** A small web page written in Python (`streamlit run app.py`) that your unit opens in a browser to see the series, the threshold, the flags or forecast with its range, and a plain-language summary.
- **Naija analogy:** The scoreboard at the stadium: everybody looks at the same screen, nobody needs to read the referee's notebook.
- **Example:** Today's notebook writes a starter `app.py` for you to finish on the server.

W6D3

🌞 Sun-angle wave (sin / cos)

- **Plain English:** A smooth yearly up-and-down added to the recipe so it knows the time of year.
- **Naija analogy:** Knowing harmattan is cold and March is hot, without being told each year.
- **Example:** `np.sin(2π × day_of_year / 365.25)` and its `cos` partner.

W3D2

🎓 Supervised vs unsupervised

- **Plain English:** A **supervised** model learns from the answer key (labels). An **unsupervised** model only sees the readings and learns what "normal" looks like. The isolation forest and the control chart are unsupervised.
- **Naija analogy:** Supervised = learning with the marking scheme. Unsupervised = learning what a normal day at the market feels like, then noticing when something is off.
- **Example:** The copycat (supervised) needs labels; the forest never sees them.

W5D1

🧪 Synthetic data

- **Plain English:** Made-up data designed to behave like the real thing. It's safe to share and safe to use with AI.
- **Naija analogy:** A mock exam: same style as WAEC, no real stakes.
- **Example:** All course datasets are synthetic, built to behave like NigComSat-1R.

W1D1

🙅🏾 Tampering

- **Plain English:** Reacting to common-cause wobble as if it were special cause. It makes things worse.
- **Naija analogy:** The driver who jams the brake for every small bump on the expressway: passengers suffer more, not less.
- **Example:** "Correcting" after every reading made the spread about 39% bigger in the simulation.

W3D2

🎯 Target

- **Plain English:** The value you expect when everything is normal. CUSUM measures how far each reading is from it.
- **Naija analogy:** The fuel gauge level you expect after buying ₦20,000 of fuel.
- **Example:** For leftovers (actual − expected), the target is 0.

W3D3

🌡️ temp_per_drive

- **Plain English:** Amplifier (TWTA) temperature divided by RF drive level: degrees per percent of drive. How hot it runs **for the work it's doing**.
- **Naija analogy:** A generator running hot while powering one bulb. The load didn't change, so something is blocking the cooling.
- **Example:** Normally 0.821 ± 0.013; during EV04 it averages 0.870, about 3.9σ high, while raw temperature never crosses its ceiling.

W4D3

🕑 The 14:00 test (look-ahead leak)

- **Plain English:** Ask of every feature: *could I compute this at 14:00 using only what I knew at 14:00?* A feature that uses the future is a **leak**: it looks brilliant in testing and is impossible in the control room.
- **Naija analogy:** Reading tomorrow's newspaper to "predict" today's football results.
- **Example:** `soc.shift(−1)` and `rolling(24, center=True)` both "knew" about EV02 at midnight, an hour before it started. They cheat.

W4D3

📐 The big-sigma trap

- **Plain English:** A feature can put a fault very far out (many σ) and still be useless, because it is also wrong on normal months it never saw.
- **Naija analogy:** A gateman who catches every thief because he stops **everybody** after October.
- **Example:** The code book's regression feature puts EV04 at 28.9σ but flags all 2,208 hours of October to December.

W4D3

🗣️ The honest forecast sentence

- **Plain English:** Most likely value, a range, and the thing that could blow it up, in one sentence.
- **Naija analogy:** Amaka on the phone from the Island.
- **Example:** "Week of 3 Nov: most likely around 56, plausibly 49 to 63; expect bigger misses at season changes."

W5D3

🔒 The leak rule (Monday-morning test)

- **Plain English:** Every feature must be something you could compute **before** the week you're forecasting begins. If not, it leaks the future and the score is fake.
- **Naija analogy:** Predicting Saturday's match result on Friday with Sunday's newspaper in your bag.
- **Example:** `shift(1)` puts 27 Oct's 74 on 3 Nov's row (fine). `shift(-1)` puts 10 Nov's 54 there (a leak).

W5D3

⏰ "The model cannot tell time"

- **Plain English:** A model like the isolation forest judges each row on its own. It doesn't know what happened an hour ago unless a column tells it.
- **Naija analogy:** A goldfish at the gate: every visitor is a stranger, even the one who came five minutes ago.
- **Example:** At 14:00 the forest sees 93% SoC, but not that it was 100% at midnight. `soc.shift(1)` gives it that memory.

W4D3

🌡️ Thermal decoupling

- **Plain English:** A component's temperature stops following the thing that normally drives it (here, RF drive).
- **Naija analogy:** Your phone getting hot while it's doing nothing.
- **Example:** EV04, cooling path degradation: the amplifier runs about 3 °C hot for its drive level for four weeks.

W4D3

🚧 Threshold line

- **Plain English:** A line drawn at the level that matters (0 dB, the SLA target), so readers instantly see breaches.
- **Naija analogy:** The red line on a fuel gauge.
- **Example:** `plt.axhline(0, color="red", ls="--")`

W1D2

📶 Throughput (Gbps)

- **Plain English:** How much data a link carries, in gigabits per second.
- **Naija analogy:** How many cars pass through the toll gate per minute.
- **Example:** Over two years, about 3.9 Gbps on an average weekday and 3.0 at weekends.

W2D2

⏱️ Time series

- **Plain English:** Measurements recorded one after another in time, where the **order matters**.
- **Naija analogy:** Your NEPA light diary: "06:00 light came, 09:00 light gone…" Shuffle it and it means nothing.
- **Example:** Daily Ka-band availability from 2023 to 2025.

W1D1

🕐 Timestamp

- **Plain English:** The date and time attached to a reading.
- **Naija analogy:** The time printed on your POS receipt.
- **Example:** `2025-08-01T21:00` = 1 August 2025, 9pm UTC.

W1D1

↔️ Tolerance

- **Plain English:** A small grace window around an event, so a flag one step early or late still counts.
- **Naija analogy:** "Service starts 10 a.m." but nobody is marked absent at 10:01.
- **Example:** `tol="1h"` for hourly files, `tol="1D"` for daily files.

W4D1

⬅️ Trailing window

- **Plain English:** A window that ends today and only looks at the past. pandas does this by default.
- **Naija analogy:** Driving using only your rear-view mirror: safe information, but always a bit late.
- **Example:** `s.rolling(30).mean()` is trailing.

W1D3

✂️ Train/test split

- **Plain English:** Dividing your data into a part the model learns from (**train**) and a part it is judged on (**test**), which it must never see while learning.
- **Naija analogy:** Past questions to practise on; the real exam to be marked on.
- **Example:** Train = January to August; test = 2 September to 31 December.

W5D1

📡 Transponder

- **Plain English:** A channel on the satellite that receives a signal and sends it back down. NigComSat-1R has 28 active ones.
- **Naija analogy:** One lane on a busy expressway. More lanes = more traffic carried.
- **Example:** `transponder_utilisation_monthly.csv` tracks how busy they are.

W1D1

📈 Trend

- **Plain English:** The long-term direction: up, down or flat.
- **Naija analogy:** Lagos population: more people every year.
- **Example:** Transponder utilisation climbing from about 50% to about 78%.

W1D1 W5D2

📈 Trend component

- **Plain English:** The slow, long-term direction once the seasonal ups and downs are averaged away.
- **Naija analogy:** Lagos traffic getting worse year after year as the city grows, whatever the day of the week.
- **Example:** Ka-band availability trend rising slowly from about 99.24% to 99.40%.

W2D1

📏 Trend-to-threshold

- **Plain English:** Fit a trend to the past, extend it into the future, and read off the day it crosses the limit.
- **Naija analogy:** "At this rate, my landlord's rent will pass my salary by 2030."
- **Example:** A straight line through three years of battery data hits 80% on 1 May 2025.

W6D2

🔑 Tuning on the answer key

- **Plain English:** Choosing a setting (a threshold, `k`, `contamination`) because it scores best on the **test** answers. The test stops being a test.
- **Naija analogy:** Adjusting your answers after the teacher reads out the marking scheme.
- **Example:** A forest budget of 0.30% gives 0 false alarms on Sep–Dec, but you'd only know that by peeking, and it misses EV03 in the past.

W5D1

🎛️ Turning the knob (tuning on the test)

- **Plain English:** Changing a setting (like z or the band width) until the test result looks good. It uses the answers, so the score is fake: expo.
- **Naija analogy:** Moving the goalpost after the ball has been kicked.
- **Example:** Raising z until all 8 test weeks fit is not a fix. The width must come from misses before the test.

W5D3

👯 Twin files

- **Plain English:** Two data files with the **same name** but different contents, in different folders.
- **Naija analogy:** Two people called "Emeka Okafor" in the same office: check the ID before you transfer money!
- **Example:** `battery_health_daily.csv` in week-03 (clean) vs week-04 (planted faults).

W2D3 W3D3

👯 Twin (redundant) features

- **Plain English:** Two columns that carry the same information (correlation above about 0.98). In a forest, which picks columns at random, a twin doubles that sensor's vote.
- **Naija analogy:** The same person voting twice with two voter's cards.
- **Example:** `soc_gap` is exactly `100 − battery_soc_pct` (correlation 1.000). The builder has 18 such pairs.

W4D3

⭐ Two stars and a wish

- **Plain English:** A way to give feedback on someone's demo: two things that were good, one thing that would make it better.
- **Naija analogy:** How a good aunty corrects you at a party: praise first, then "but next time, my dear…".
- **Example:** The peer scoring sheet has a box for each star and the wish.

W6D3

🌡️ TWTA and RF drive

- **Plain English:** The TWTA (travelling-wave tube amplifier) boosts the signal before it's sent down. RF drive is how hard it's being pushed.
- **Naija analogy:** A generator: the harder you load it, the hotter it runs.
- **Example:** `twta_thermal_daily.csv` : temperature vs drive, r = 0.59 on levels, 0.40 on changes.

W2D2

📣 TWTA (travelling-wave tube amplifier)

- **Plain English:** The satellite part that makes our signal strong enough to reach the ground. It runs hot.
- **Naija analogy:** The loudspeaker at a party: bigger volume, more heat, and you notice when it starts overheating.
- **Example:** `twta_thermal_daily.csv` tracks its daily temperature.

W3D3

🏢 Unit head

- **Plain English:** The head of your department. They agree your project's scope by email.
- **Naija analogy:** Your HOD signing your project approval form.
- **Example:** Their reply email goes into your final submission.

W2D3

🧭 Unseen months

- **Plain English:** Data the model or feature was not fitted on. The honest place to check whether it still behaves.
- **Naija analogy:** A new driver who has only driven in the dry season. Wait until he meets the July rain on Third Mainland Bridge.
- **Example:** Fitted on January to May, the residual drifts from June and breaks completely from October.

W4D3

🐶 Unsupervised

- **Plain English:** Learning without an answer key. The model never sees the labels; it only learns what is common.
- **Naija analogy:** Learning which neighbours are "normal" just by living on the street for a year, without anyone telling you who is a thief.
- **Example:** `fit()` gets X only. The labels are used afterwards, to **score** it.

W4D2

🌧️ Unusual ≠ anomaly

- **Plain English:** A reading can be statistically rare yet perfectly explained by known physics. Not everything rare is a fault.
- **Naija analogy:** Heavy traffic on Third Mainland Bridge at 6 p.m. on a Friday is unusual compared with 3 a.m., but nobody calls it an emergency.
- **Example:** Rain fade drops C/N below the naive line, but only 12 of the 18 flagged hours were real outages.

W4D1

📡 Uplink power

- **Plain English:** How strongly the ground station transmits up to the satellite. It's turned up automatically when rain weakens the signal.
- **Naija analogy:** Shouting louder when the generator next door starts. You shout *because* of the noise.
- **Example:** Uplink power averages about 41 dBm when dry and about 44 dBm in rain.

W1D2

🎚️ Uplink power control

- **Plain English:** The ground station automatically raises its transmit power when rain weakens the signal.
- **Naija analogy:** Talking louder on the phone when the network starts breaking.
- **Example:** `uplink_power_dbm` rises in heavy rain: a response, not a cause.

W2D2

🌍 UTC

- **Plain English:** The world's reference clock. Satellites and telemetry usually record in UTC.
- **Naija analogy:** The "master clock" everyone agrees on, like full-time at Old Trafford: the same moment for the whole world.
- **Example:** The eclipse happens at 21:00 UTC.

W1D1

🚶 Walk-forward backtest (expanding window)

- **Plain English:** Cut the data at a point, train on everything before it, forecast the next few weeks, score them; then move the cut forward and do it again. The training block only grows.
- **Naija analogy:** Pretend it's June, make your call, check it; pretend it's July, make your call, check it. No time travel allowed.
- **Example:** `walk_forward_mae(tickets, make_features, X_COLS, step=4, h=4)` gives six fold MAEs: 2.46, 4.67, 2.81, 1.27, 7.67, 5.15.

W6D1

🚶 Walk-forward misses (honest past errors)

- **Plain English:** For each past week, fit only on the weeks before it, forecast that week, and write down the miss. These are the misses the model really made before knowing the answer.
- **Naija analogy:** Checking a tipster's record on matches he predicted **before** kick-off, not after.
- **Example:** 32 walk-forward misses from 24 Mar 2025; the biggest, +15.1 on 7 Apr, is the week the rains arrived.

W5D3

🕳️ Warm-up NaN

- **Plain English:** The empty values at the start of a rolling result, before the window has enough rows.
- **Naija analogy:** You can't average 7 days of salary when you only started work 3 days ago.
- **Example:** `rolling(7)` gives 6 `NaN` values at the start.

W1D3

🚧 Warning line

- **Plain English:** A limit you set on a speed or rate (not on the level) that triggers a check.
- **Naija analogy:** The speed limit sign: you can still be far from your destination and be going too fast.
- **Example:** −3.0 Ah per year on the battery's rolling slope.

W3D3

⏳ Warning time (lead time)

- **Plain English:** How long you have to act between making the call and the crossing. More data gives a closer call but less time to act.
- **Naija analogy:** Knowing about the exam a month ahead versus the night before.
- **Example:** 1,229 days of warning with 1 year of data; only 134 with 4.

W6D2

🇳🇬 WAT

- **Plain English:** West Africa Time, Nigeria's clock. Always **UTC + 1**, with no daylight saving.
- **Naija analogy:** Lagos time is UTC time plus one hour. (“African time” is another matter 😄)
- **Example:** 21:00 UTC = 22:00 WAT.

W1D1

🕰️ Week-of-year clock (`sin52` , `cos52`)

- **Plain English:** Two columns that place each week on a circle, so the model sees the yearly cycle and week 52 sits next to week 1.
- **Naija analogy:** The hour hand: 11 o'clock and 1 o'clock are close, even though 11 and 1 are far apart as numbers.
- **Example:** `np.sin(2 * np.pi * woy / 52)` and `np.cos(...)`. A circle is smooth, so it can't make the rainy season's sudden on/off step.

W5D3

🎶 Weekly rhythm (day-of-week effect)

- **Plain English:** A pattern that repeats every 7 days because people work Monday to Friday.
- **Naija analogy:** Lagos traffic: wahala on weekdays, free roads on Sunday morning.
- **Example:** Weekdays average about 7 tickets, weekends about 3.4.

W1D3

🕵🏾 "Where did that number come from?"

- **Plain English:** The question to ask of every alarm limit. A good limit has a reason you can write down.
- **Naija analogy:** "Who told you the price is ₦5,000?" before you pay at the market.
- **Example:** 20 A was a guess; mean − 3σ of sunlit data (about 26.4 A) has a reason.

W3D1

↔️ Wide format

- **Plain English:** One row per time, one column per group. What charts want.
- **Naija analogy:** The same register as a grid: students across the top, days down the side.
- **Example:** `wide` has 52 rows and 6 zone columns.

W1D2

📏 Window size

- **Plain English:** How many rows go into each calculation. Bigger = smoother but slower.
- **Naija analogy:** How many days of gossip you collect before you believe a rumour.
- **Example:** 3, 7 and 30-day windows on the same ticket data.

W1D3

🧱 X (the feature matrix)

- **Plain English:** The table of numbers you feed a model. **Rows are the things you judge (hours), columns are what you know about each one (channels).**
- **Naija analogy:** The class register: one row per student, one column per subject score.
- **Example:** `X = bus[CHANNELS]` : 8,760 hours × 6 channels.

W4D2

📏 Y-axis range

- **Plain English:** Where the vertical axis starts and ends. Counts should start at 0; percentages that live near the top may zoom in, if you say so on the chart.
- **Naija analogy:** A "before and after" photo zoomed onto the waist: fair if you say it's zoomed. 😄
- **Example:** Availability on 96–100%; tickets per weekday from 0.

W1D2

🚦 Yellow and red limits

- **Plain English:** Two levels: yellow = warning, red = act now.
- **Naija analogy:** The fuel light comes on (yellow), then the car stops on Third Mainland Bridge (red).
- **Example:** Ground-station software often has a yellow and a red limit on each telemetry channel.

W3D1

🔢 z-score (explaining a flag)

- **Plain English:** How many standard deviations a reading is from its normal. Per channel, it tells you **which** channel made the hour odd.
- **Naija analogy:** "Your blood pressure is fine, your sugar is fine, but your temperature is 40." The doctor points at the one number that's off.
- **Example:** EV02's first hour: state of charge −20.7σ. That's the sentence for the duty engineer.

W4D2